

Yukawa Quantum Field Theory in Two Space-Time Dimensions Without Cutoffs*

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This paper provides estimates, which permit the construction of the physical vacuum and the corresponding Hilbert space of the Y_2 theory. The representation of the local field algebra thus obtained is locally Fock.

One estimate is a volume-dependent first-order N_τ estimate, which implies in particular that the vacuum energy per unit volume is bounded from below.

The second estimate states that the vacuum energy per unit volume is bounded from above.

1. INTRODUCTION

The discussion of the Yukawa quantum field theory in two space-time dimensions has up until now been studied as a Hamiltonian theory in Fock space. This was possible due to the fact that a space cut off $g \in \mathfrak{S}(\mathbb{R})$ was used in the definition of the Hamiltonian $H(g)$ in order to violate one of the assumptions (translation invariance) leading to Haag's theorem.

This Hamiltonian is formally [9]

$$H(g) = H_0 + H_I(g) + c(g), \quad (1.1)$$

where

$$H_I(g) = \lambda \int g(x) \varphi(x) : \bar{\psi}(x) \psi(x) : dx \quad (1.2)$$

describes the interaction between a bose field φ and a fermi field ψ . $c(g)$ is an infinite counterterm defined by second-order perturbation theory.

A first step to control $H(g)$ as $g \rightarrow 1$ is to look for the behavior of its ground-state energy $E(g)$. A second step is to improve the well-known first-order N_τ estimate [6, 11] for fixed g

$$0 \leq N_\tau \leq c_1(H(g) + c_2)(0 \leq \tau < 1, c_1 = c_1(\tau), c_2 = c_2(g, \tau)) \quad (1.3)$$

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to an estimate which is uniform in g . We will show the following: Let $\mathfrak{G}(\alpha)$ ($\alpha > 0$) be set of all $g \in C_0^\infty(\mathbb{R})$ having the following properties

- (a) $0 \leq g \leq 1$,
- (b) $\sup_{\substack{m \leq 6 \\ x \in \mathbb{R}}} |\partial^m g(x)/\partial x^m| \leq \alpha$.

Set

$$m(g) = \text{measure}(\text{supp } g). \quad (1.4)$$

Then we have the

THEOREM 1.1. *For given τ , α ($0 \leq \tau < 1$, $\alpha > 0$) there exist $a = a(\tau, \alpha) > 0$ and $C = C(\tau)$ such that*

$$E(g) \leq am(g), \quad (1.5)$$

$$0 \leq N_\tau \leq C(H(g) + m(g)a), \quad (1.6)$$

for all $g \in \mathfrak{G}(\alpha)$.

This theorem remains true if we add a boson self-interaction term of the form

$$:P(\varphi)(g): = \sum_{n=0}^{2m} \int a_n : \varphi(x)^n : g(x) dx, \quad (a_n \text{ real}, a_{2m} > 0) \quad (1.7)$$

to the Hamiltonian [16]. Indeed the estimate

$$:P(\varphi)(g): \geq -\epsilon N + a' m(g) \quad (\epsilon > 0, a' = a'(\alpha), g \in \mathfrak{G}(\alpha)) \quad (1.8)$$

has been established in [8], so the addition of (1.8) to (1.6) gives the estimate (1.6) for the new Hamiltonian.

Estimate (1.5) for the new Hamiltonian will follow from the proof of (1.5).

Now let

$$\omega_g(\cdot) = (\Phi_g, \cdot \Phi_g) \quad (1.9)$$

be the ground-state functional of a ground state Φ_g of $H(g)$. [$E(g)$ is not known to be a simple eigenvalue of $H(g)$.] Then we have the immediate

COROLLARY 1.2. *Given $\alpha > 0$ and τ ($0 \leq \tau < 1$) there exists $d = d(\alpha, \tau) < \infty$ such that*

$$\omega_g(N_\tau) \leq dm(g) \quad (1.10)$$

uniformly for all $g \in \mathfrak{G}(\alpha)$.

Estimate (1.10) will ensure the local Fock property of the physical representations. To prove the estimate (1.6), we will use localization indices introduced by J. Glim and A. Jaffe [8] showing that the ground-state energy per unit volume of the $(\varphi^4)_2$ theory is finite.

We also establish some localized N_τ estimates in Chapter 2. In Chapter 3 we prove certain estimates on kernels of localized Wick-ordered monomials. The combination of these results will then allow us to prove the estimate (1.6) in Chapter 4. The proof of estimate (1.5) is relatively easy, if we use the unitary dressing transformation suggested by P. Federbush and B. Gidas [3], and will be given in Chapter 5. We note that the corresponding estimate in the $(\varphi^{2n})_2$ theory is trivial, since the free vacuum is in the domain of $H(g)^{1/2}$ and hence the vacuum-energy renormalization constant is finite, yielding $E(g) \leq 0$ for the Wick-ordered Hamiltonian. Chapter 6 gives the outline of the proof of the local Fock property of the physical representation.

To make the essential steps in the proof more transparent, we will make the following simplification. Let $h \in C_0^\infty(\mathbb{R})$ ($0 \leq h \leq 1$) be such that

$$h_j(x) = h(x - j), j \in \mathbb{Z} \quad (1.11)$$

defines a decomposition of unity

$$\sum_{j \in \mathbb{Z}} h_j(x) = 1 \quad \text{for all } x \in \mathbb{R}. \quad (1.12)$$

Note that there exists $\beta = \beta(h) > 0$ such that $h_j \in \mathfrak{G}(\beta)$ for all j . For an arbitrary $g \in \mathfrak{G}(\alpha)$ set $J(g) = \{j \in \mathbb{Z} \mid \text{supp } h_j \cap \text{supp } g \neq \emptyset\}$ such that $|J(g)| \leq m(g) + 2$, where $|J(g)|$ is the number of elements in $J(g)$. Then

$$g = \sum_{j \in J(g)} h_j g \quad (1.13)$$

and $h_j g \in \mathfrak{G}(\alpha\beta)$ for all j . Now instead of studying $H(g)$ ($g \in \mathfrak{G}(\alpha)$) we will simply consider the family $H(g_n)$ ($n \in \mathbb{Z}^+$) where

$$g_n = \sum_{|j| \leq n} h_j, \quad (1.14)$$

and where h is fixed and satisfies (1.12). Note that $2n - 2 \leq m(g_n) \leq 2n + 2$. An inspection of the proof will show that this is no substantial simplification. Finally our notation will be chosen as close as possible to the one used in the Les Houches lecture notes of J. Glimm and A. Jaffe [11]. Also the reader is supposed to be familiar with the proof of (1.3) presented there.

2. LOCAL N_τ ESTIMATES

In this chapter we will show how Wick monomials, whose kernels have certain support properties in configuration space, may be estimated with help of local N_τ operators. Since we also want to take the sum of such Wick monomials, we will also give sufficient conditions for estimating the sum of local N_τ operators by (global) N_τ operators.

For given momentum $k \in \mathbb{R}$,

$$\begin{aligned}\mu(k, 0) &= (k^2 + m_b^2)^{1/2}, \\ \mu(k, \pm 1) &= (k^2 + m_f^2)^{1/2},\end{aligned}\tag{2.1}$$

denotes the energy of a boson resp. fermion.

Let $b(k, \epsilon)$ and $b(k, \epsilon)^*$ ($\epsilon = 0, \pm 1$) be the annihilation and creation operators for bosons ($\epsilon = 0$) and fermions ($\epsilon = \pm 1$) with momentum k in the Fock space $\mathfrak{F}$ of our theory. Define the following annihilation and creation operators at the space point x :

$$\begin{aligned}b(x, \epsilon) &= (2\pi)^{-1/2} \int e^{ikx} b(k, \epsilon) dk, \\ b(x, \epsilon)^* &= (2\pi)^{-1/2} \int e^{-ikx} b(k, \epsilon)^* dk.\end{aligned}\tag{2.2}$$

Let $\mu_x^\tau(\epsilon)$ with kernel $k_\tau(\cdot, \epsilon) \in O_C'(\mathbb{R})$ be the Fourier transform of the multiplication operator $\mu^\tau(\cdot, \epsilon) \in O_M(\mathbb{R})$ on $\mathfrak{S}(\mathbb{R})$ in momentum space (for notation see [17]). For given $\zeta \in O_M(\mathbb{R})$ (ζ real) and $\tau \in \mathbb{R}$, define $N_{\tau, \zeta}^\epsilon$ to be the biquantization of $\zeta \mu_x^\tau(\epsilon) \zeta$ in the sense of [8]. Formally we have

$$N_{\tau, \zeta}^\epsilon = \int b(x, \epsilon)^* \zeta(x) k_\tau(x - y, \epsilon) \zeta(y) b(y, \epsilon) dx dy.\tag{2.3}$$

More precisely $N_{\tau, \zeta}^\epsilon$ is the Fiedrichs extension of the positive operator defined by the right-hand side of (2.3) on D , the subspace of vectors in Fock space with a finite number of particles and wave functions in the Schwartz spaces $\mathfrak{S}(\mathbb{R}^n)$. $N_{\tau, \zeta}^\epsilon$ is defined on D since $\zeta \mu_x^\tau(\epsilon) \zeta$ is a bounded linear operator from $\mathfrak{S}(\mathbb{R})$ to $L^2(\mathbb{R})$.

We will be interested in the estimate of Wick monomials of the form

$$W = W(\epsilon) = \int A(\mathbf{x}, \epsilon) w(\mathbf{x}) d\mathbf{x}\tag{2.4}$$

with

$$A(\mathbf{x}, \epsilon) = \prod_{i=1}^r b(x_i, \epsilon_i)^* \prod_{i=r+1}^{r+s} b(x_i, \epsilon_i), \quad (\mathbf{x}, \epsilon) = (x_1, \epsilon_1; x_2, \epsilon_2; \dots; x_{r+s}, \epsilon_{r+s}), \quad (2.5)$$

where $w(\mathbf{x}) \in \mathfrak{S}'(\mathbb{R}^n)$ has certain support properties.

For this purpose define

$$(\mu^{\tau/2} \zeta b)(x, \epsilon) = \int k_{\tau/2}(x - y, \epsilon) \zeta(y) b(y) dy, \quad \zeta \in O_M(\mathbb{R}) \text{ } (\zeta \text{ real}), \tau \in \mathbb{R}. \quad (2.6)$$

The “annihilation” operators $b(x, \epsilon)$ and $(\mu^{\tau/2} \zeta b)(x, \epsilon)$ map D into D and depend on the parameter x . If we regard $x \in \mathbb{R}$ as a variable and not as a parameter, then

$$\begin{aligned} b(\cdot, \epsilon) : D &\rightarrow S(\mathbb{R}^s) \otimes D \subset L^2(\mathbb{R}^s) \otimes D, \\ (\mu^{\tau/2} \zeta b)(\cdot, \epsilon) : D &\rightarrow S(\mathbb{R}^s) \otimes D \subset L^2(\mathbb{R}^s) \otimes D. \end{aligned} \quad (2.7)$$

Here $S(\mathbb{R}^s) \otimes D$ is the subspace consisting of vectors with a finite number of particles (at least s , the first s variables not being symmetrized) and wave functions in the corresponding Schwartz spaces $\mathfrak{S}(\mathbb{R}^n)$. Consider the operator

$$A(\mathbf{x}, \epsilon) = \prod_{i=1}^p b(x_i, \epsilon_i) \prod_{i=p+1}^{p+q} (\mu^{\tau_i/2} \zeta_i b)(x_i, \epsilon_i), \quad (\zeta_i \in O_M(\mathbb{R}), \zeta_i \text{ real}, i \geq p+1) \quad (2.8)$$

such that

$$A(\cdot, \epsilon) : D \rightarrow S(\mathbb{R}^{p+q}) \otimes D \subset L^2(\mathbb{R}^{p+q}) \otimes D. \quad (2.9)$$

LEMMA 2.1. *Suppose the operator $A(\cdot, \epsilon)$ defined by (2.8) satisfies the following condition: For all i, j ($i, j > p, i \neq j$) either $\text{supp } \zeta_i \cap \text{supp } \zeta_j = \emptyset$ or $\epsilon_i \neq \epsilon_j$.*

Then the operator

$$B(\cdot, \epsilon) = A(\cdot, \epsilon)(N+1)^{-p/2} \prod_{i=p+1}^{p+q} (N_{\tau_i, \zeta_i}^{\epsilon_i} + 1)^{-1/2} \quad (2.10)$$

satisfies

$$\|B(\cdot, \epsilon)\| \leq 1. \quad (2.11)$$

Note that by the assumption of the lemma $[N_{\tau_i, \zeta_i}^{\epsilon_i}, N_{\tau_j, \zeta_j}^{\epsilon_j}] = 0$ for all $i, j > p$.

Proof. We have the following relations on D

$$\begin{aligned} b(x, \epsilon)(N+n)^{-1/2} &= (N+n+1)^{-1/2} b(x, \epsilon), \quad n \in \mathbb{Z}^+ \\ (\mu^{\tau/2} \zeta b)(x, \epsilon)(N+n)^{-1/2} &= (N+n+1)^{-1/2} (\mu^{\tau/2} \zeta b)(x, \epsilon), \quad \zeta \in O_M(\mathbb{R}) (\zeta \text{ real}), \tau \in \mathbb{R} \end{aligned}$$

and by the assumption of the lemma

$$(\mu^{\tau_i/2} \zeta_i b)(x_i, \epsilon_i) N_{\tau_j, \zeta_j}^{\epsilon_j} = N_{\tau_j, \zeta_j}^{\epsilon_j} (\mu^{\tau_i/2} \zeta_i b)(x_i, \epsilon_i), \quad i \neq j.$$

This permits us to write $B(\cdot, \epsilon)$ as a product of $p + q$ operators mapping

$$S(\mathbb{R}^j) \otimes D \text{ into } S(\mathbb{R}^{j+1}) \otimes D \quad (0 \leq j \leq p + q - 1).$$

The estimates

$$\|b(\cdot, \epsilon) \Phi\| \leq \|N^{1/2} \Phi\|, \quad (2.12)$$

$$\|(\mu^{\tau/2} \zeta b)(\cdot, \epsilon) \Phi\| \leq \|(N_{\tau, \zeta}^{\epsilon})^{1/2} \Phi\|, \quad \Phi \in \text{Fock space}, \quad (2.13)$$

then show that each operator is bounded in norm by 1.

Now (2.12) is proved in [11] and (2.13) is established similarly

$$\begin{aligned} \|(\mu^{\tau/2} \zeta b)(\cdot, \epsilon) \Phi\|^2 &= \int (\Phi, (\mu^{\tau/2} \zeta b)(x, \epsilon)^* (\mu^{\tau/2} \zeta b)(x, \epsilon) \Phi) dx \\ &= (\Phi, N_{\tau, \zeta}^{\epsilon} \Phi) = \|(N_{\tau, \zeta}^{\epsilon})^{1/2} \Phi\|^2 \quad \Phi \in \text{Fock space}. \end{aligned} \quad (2.14)$$

Let us return to our operator of the form (2.4). We will be interested in the following case. Let

$$A \cup B \cup C = \{1, 2, \dots, r + s\}$$

be a disjoint union such that $C \subset \{1, \dots, r\}$, $A \subset \{r + 1, \dots, r + s\}$ and such that there exist real $\zeta_i \in O_M(\mathbb{R})$ ($i \in A \cup C$) with the following properties

$$\text{supp } \zeta_i \cap \text{supp } \zeta_j = \emptyset \quad \text{or} \quad \epsilon_i \neq \epsilon_j \quad \text{for} \quad i, j \in A \quad \text{or} \quad i, j \in C, i \neq j \quad (2.15)$$

and

$$w(\mathbf{x}) = \prod_{i \in A \cup C} \zeta_i(x_i) w(\mathbf{x}) \quad \text{for all } \mathbf{x} \in \mathbb{R}^{r+s}. \quad (2.16)$$

Let $\tilde{w}(\mathbf{k})$ ($\mathbf{k} = (k_1, \dots, k_{r+s})$) be the Fourier transform of $w(\mathbf{x})$. Then $\prod_{i \in A \cup C} \mu_i^{-\tau_i/2} w$ is defined to be the Fourier transform of $\prod_{i \in A \cup C} \mu_i^{-\tau_i/2} (k_i, \epsilon_i) \tilde{w}(\mathbf{k})$.

We have

LEMMA 2.2. *Assume (2.16) holds for a certain choice of real $\zeta_i \in O_M(\mathbb{R})$ ($i \in A \cup C$) satisfying (2.15). Furthermore assume also that for some τ_i ($i \in A \cup C$) $\prod_{i \in A \cup C} \mu_i^{-\tau_i/2} w$ defines a bounded linear operator from $L^2(\mathbb{R}^s) \otimes D$ into $L^2(\mathbb{R}^r) \otimes D$.*

Then

$$\left\| \prod_{i \in C} (N_{\tau_i, \zeta_i}^{\epsilon_i} + 1)^{-1/2} W(N + 1)^{-|B|/2} \prod_{i \in A} (N_{\tau_i, \zeta_i}^{\epsilon_i} + 1)^{-1/2} \right\| \leq \text{const} \left\| \prod_{i \in A \cup C} \mu_i^{-\tau_i/2} w \right\| \quad (2.17)$$

with the constant less than $O(|r - s| + 1)^{|B|/2}$.

Proof. Due to (2.16) we have

$$W = \int \prod_{i \in C} (\mu^{\tau_i/2} \zeta_i b)(x_i, \epsilon_i)^* \prod_{\substack{i \in B \\ i \leq r}} b(x_i, \epsilon_i)^* \prod_{\substack{i \in B \\ i > r}} b(x_i, \epsilon_i) \prod_{i \in A} (\mu^{\tau_i/2} \zeta_i b)(x_i, \epsilon_i) \\ \times \left(\prod_{i \in A \cup C} \mu^{-\tau_i/2} w \right) (\mathbf{x}) d\mathbf{x} \quad (2.18)$$

and (2.17) follows in the usual way from Lemma 2.1 (see e.g. [11]). Note that factors $(N+1)^{-1/2}$ may be placed on either side of W .

Remark 2.3. If $\prod_{i \in A \cup C} \mu_i^{-\tau_i/2} w \in L^2(\mathbb{R}^{r+s})$, then $\prod_{i \in A \cup C} \mu_i^{-\tau_i/2} w$ defines a bounded linear operator from $L^2(\mathbb{R}^s) \otimes D$ into $L^2(\mathbb{R}^r) \otimes D$ and

$$\left\| \prod_{i \in A \cup C} \mu_i^{-\tau_i/2} w \right\| \leq \left\| \prod_{i \in A \cup C} \mu_i^{-\tau_i/2} w \right\|_2. \quad (2.19)$$

LEMMA 2.4. *Let the hypotheses be as in Lemma 2.2 with $\prod_{i \in A \cup C} \mu_i^{-\tau_i/2} w \in L^2(\mathbb{R}^{r+s})$ and assume in addition that B contains at least one fermion variable. Then*

$$\left\| \prod_{i \in C} (N_{\tau_i, \zeta_i}^{\epsilon_i} + 1)^{-1/2} W (N+1)^{-(|B|-1)/2} \prod_{i \in A} (N_{\tau_i, \zeta_i}^{\epsilon_i} + 1)^{-1/2} \right\| \\ \leq \text{const} \left\| \prod_{i \in A \cup C} \mu_i^{-\tau_i/2} w \right\|_2. \quad (2.20)$$

The proof again follows from Lemma 2.1 and a standard estimate on fermion operators (see [11, Prop. 1.2.3.b]).

In the above estimates, we were allowed to place factors $(N+1)^{-1/2}$ on either side of W . The problem when this is allowed for factors of the form $(N_{0, \zeta_i} + 1)^{-1/2}$ is answered by the following lemma.

LEMMA 2.5. *Let the hypotheses be as in Lemma 2.2 with $|B| = 0$. Let $I = I_1 \cup I_2$ ($I_1 \cap I_2 = \emptyset$) be a nonempty subset of $\{1, \dots, r+s\}$ with the following properties*

- (1) $\tau_i = 0$ for $i \in I$;
- (2) $\text{supp } \zeta_i \cap \text{supp } \zeta_j = \emptyset$ or $\epsilon_i \neq \epsilon_j$ for all $i \in I$ and all $j \neq i$.

Set

$$I_1' = \{i \mid i \in I_1 \text{ or } i \leq r, i \notin I_2\} \supset I_1,$$

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$$I_2' = \{i \mid i \in I_2 \text{ or } i > r, i \notin I_1\} \supset I_2,$$

such that $I_1' \cup I_2' = \{1, \dots, r+s\}$ is a disjoint union. Then

$$\left\| \prod_{i \in I_1'} (N_{\tau_i, \zeta_i}^{\epsilon_i} + 1)^{-1/2} W \prod_{i \in I_2'} (N_{\tau_i, \zeta_i}^{\epsilon_i} + 1)^{-1/2} \right\| \leq \text{const} \left\| \prod_{i=1}^{r+s} \mu_i^{-\tau_i/2} w \right\|. \quad (2.21)$$

The proof is again simple due to the fact that by assumption

$$(N_{0,\zeta_i}^{\epsilon_i} + a)^{-1/2} W = W(N_{0,\zeta_i}^{\epsilon_i} + a \pm 1)^{-1/2}, \quad a > 1, \quad i \in I \quad (2.22)$$

with plus sign for $i \leq r$ and minus sign for $i > r$.

Remark 2.6. If I contains a fermion index, i_0 say (i.e., $\epsilon_{i_0} \neq 0$), and if $\prod_{i=1}^{r+s} \mu_i^{-\tau_i/2} w \in L^2(\mathbb{R}^{r+s})$, then (2.21) is still true if we delete the factor $(N_{0,\zeta_{i_0}}^{\epsilon_{i_0}} + 1)^{-1/2}$ from the left-hand side of (2.21) and replace the right-hand side by the L^2 -norm of $\prod_{i=1}^{r+s} \mu_i^{-\tau_i/2} w$.

Now we will assume that the kernel w is smooth, i.e., $w \in \mathfrak{S}(\mathbb{R}^{r+s})$. In the case of ordinary estimates, it is known that only the boson-operator part has to be bound by a power of the number operator. When w has certain support properties we have the following local N_τ estimate as an analogue:

LEMMA 2.7. *Let $B = \{i \mid \epsilon_i = 0\}$, $F = \{i \mid \epsilon_i \neq 0\}$ be the decomposition of $(1 \cdots r + s)$ into boson and fermion indices. Assume there exist $\zeta_i \in O_M(\mathbb{R}) (i \in B)$ with $\text{supp } \zeta_i \cap \text{supp } \zeta_j = \emptyset$ ($i \neq j, i, j \in B$) such that $w(\mathbf{x}) = \prod_{i \in B} \zeta_i(x_i) w(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^{r+s}$. Then there exists a $\mathfrak{S}(\mathbb{R}^{r+s})$ -norm s such that*

$$\left\| \prod_{i \in B} (N_{0,\zeta_i}^0 + 1)^{-1/2} W \right\| \leq \text{const } \|w\|_s. \quad (2.23)$$

Remark 2.8. Inspection of the proof of Proposition 1.2.3c in [11] shows that the s -norm in (2.23) may be chosen translationally invariant, i.e., $\|w\|_s = \|w_{\mathbf{a}}\|_s$ for all $\mathbf{a} \in \mathbb{R}^{r+s}$ with $w_{\mathbf{a}}(\mathbf{x}) = w(\mathbf{x} - \mathbf{a})$. This follows easily from the fact that $\tilde{w}_{\mathbf{a}}(\mathbf{k}) = e^{i\mathbf{k}\mathbf{a}} \tilde{w}(\mathbf{k}) = \prod_{i=1}^{r+s} e^{ik_i a_i} \tilde{w}(\mathbf{k})$.

Finally we need a lemma in the case of overlapping support. Since we do not need the most general case, we will be satisfied with the following

LEMMA 2.9. *Assume there exist real $\zeta_i \in O_M(\mathbb{R}) (i = 1 \cdots r + s)$, which are uniformly bounded, such that $\prod_{i=1}^{r+s} \zeta_i(x_i) w(\mathbf{x}) = w(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^{r+s}$. Then*

$$\left\| W \prod_{i=1}^{r+s} (N_{0,\zeta_i}^{\epsilon_i} + 1)^{-1/2} \right\| \leq \text{const } \|w\|. \quad (2.24)$$

Each of the factors $(N_{0,\zeta_i}^{\epsilon_i} + 1)^{-1/2}$ may be placed on either side of W .

Proof. We have

$$(\zeta_i b)(x_i, \epsilon_i) N_{0,\zeta_j}^{\epsilon_j} = (N_{0,\zeta_j}^{\epsilon_j} + \delta_{\epsilon_i, \epsilon_j} \zeta_j^2(x_i)) (\zeta_i b)(x_i, \epsilon_i).$$

But then also

$$(\zeta_i b)(x_i, \epsilon_i)(N_{0, \zeta_i}^{\epsilon_i} + a)^{-1/2} = (N_{0, \zeta_i}^{\epsilon_i} + \delta_{\epsilon_i, \epsilon_j} \zeta_j^2(x_i) + a)^{-1/2} (\zeta_i b)(x_i, \epsilon_i) \quad (a > 0)$$

by which, e.g.,

$$\left\| \prod_{i=1}^r (\zeta_i b)(\cdot, \epsilon_i) \prod_{i=1}^r (N_{0, \zeta_i}^{\epsilon_i} + 1)^{-1/2} \right\| \leq 1$$

as above [see (2.11)] and hence

$$\left\| \prod_{i=1}^r (N_{0, \zeta_i}^{\epsilon_i} + 1)^{-1/2} W \prod_{i=r+1}^s (N_{0, \zeta_i}^{\epsilon_i} + 1)^{-1/2} \right\| \leq \text{const } \|w\|. \quad (2.25)$$

Since the ζ_i are uniformly bounded, we may move the first r factors $(N_{0, \zeta_i}^{\epsilon_i} + 1)^{-1/2}$ to the right, proving (2.24). The last statement of the lemma follows from the same argument.

This completes the discussion of local N_τ estimates. The next problem we will be concerned with is the question of summability of local N_τ operators. For this we need two preparatory lemmas. By definition the kernel $k_\tau(x, \epsilon)$ of $\mu_x^\tau(\epsilon) \in O_{C'}(\mathbb{R})$ is

$$k_\tau(x, \epsilon) = (2\pi)^{-1/2} \int e^{-ikx} \mu^\tau(k, \epsilon) dk. \quad (2.26)$$

We will only consider the case $0 < \tau < 1$ and hence in particular (see e.g. [19 p. 185])

$$k_\tau(x, \epsilon) = \left[2^{\tau/2+1} / \Gamma\left(-\frac{\tau}{2}\right) \right] \left(\frac{m}{|x|} \right)^{(\tau+1)/2} \int_0^\infty e^{-m|x|\cosh t} \cosh t \left(\frac{1+\tau}{2} \right) dt, \quad (2.27)$$

where $m = m_b$ (resp m_f) if $\epsilon = 0$ (resp ± 1). Equation (2.27) gives the asymptotic behavior

$$|(\partial^n / \partial x^n) k_\tau(x, \epsilon)| \leq O(e^{-m|x|}) \text{ as } |x| \rightarrow \infty. \quad (2.28)$$

Furthermore for any $\delta > 0$

$$|k_\tau(x, \epsilon)| \leq C(\delta) |x|^{-1-\tau-\delta} \text{ for all } x. \quad (2.29)$$

To prove (2.29) we note that

$$x^2 k_\tau(x, \epsilon) = -(2\pi)^{-1/2} \int e^{-ikx} \frac{\partial^2}{\partial k^2} \mu^\tau(k, \epsilon) dk. \quad (2.30)$$

Since

$$|(\partial^n / \partial k^n) \mu^\tau(k, \epsilon)| \leq \text{const } \mu^{-n+\tau}(k, \epsilon)$$

we have $x^2 k_\tau(x, \epsilon) \in L^\infty(\mathbb{R})$ and $x^2 k_\tau(x, \epsilon)|_{x=0} = 0$. Hence

$$\begin{aligned} |x^2 k_\tau(x, \epsilon)| &= (2\pi)^{-1/2} \left| \int (e^{-ikx} - 1) \frac{\partial^2}{\partial k^2} \mu^\tau(k, \epsilon) dk \right| \\ &\leq \frac{2|x|^\Theta}{(2\pi)^{1/2}} \int |k|^\Theta \left| \frac{\partial^2}{\partial k^2} \mu^\tau(k, \epsilon) \right| dk \\ &\leq C(\Theta) |x|^\Theta \quad \text{for } 0 < \Theta < 1 - \tau \end{aligned}$$

proving (2.29). In particular the estimate (2.29) gives for $h_\tau(x, \epsilon) = x k_\tau(x, \epsilon)$:

LEMMA 2.10. *For $0 < \tau < 1$, $h_\tau(\cdot, \epsilon)$ is in $L^1(\mathbb{R})$.*

As an application, we have the following

LEMMA 2.11. *Let the set $\{\zeta_l\}_{l \in \mathbb{Z}} \subset O_M(\mathbb{R})$ of real functions satisfy the following two conditions*

- (i) $\sup_x \sum_l |\zeta_l(x)| \leq \alpha$, $\sup_x \sum_l \zeta_l^2(x) \leq \beta^2 - 1$, for some $\alpha > 0$, $\beta > 1$ (in particular the ζ_l are uniformly bounded);
- (ii) $\sup_l |\zeta_l(x) - \zeta_l(y)| \leq \gamma |x - y|$ for all $x, y \in \mathbb{R}$ and some $\gamma > 0$.

Then, as a relation between bilinear forms on $\mathfrak{S}(\mathbb{R}) \times \mathfrak{S}(\mathbb{R})$,

$$\sum_{l \in \mathbb{Z}} \zeta_l \mu_{x^\tau}(\epsilon) \zeta_l \leq \text{const } \mu_{x^\tau}(\epsilon) \quad (0 \leq \tau < 1), \quad (2.31)$$

where the constant depends on τ , α , β and γ .

Proof. For $\tau = 0$ this is trivial, so we will assume $0 < \tau < 1$. Let $\psi \in \mathfrak{S}(\mathbb{R})$ and denote by $(\cdot, \cdot)$ the scalar product in $L^2(\mathbb{R})$. We write $0 \leq (\psi, \zeta_l \mu_{x^\tau}(\epsilon) \zeta_l \psi) = (\psi, \mu_{x^\tau} \zeta_l^2 \psi) + (\psi, \zeta_l [\mu_{x^\tau}(\epsilon), \zeta_l] \psi)$.

Now

$$([\mu_{x^\tau}(\epsilon), \zeta_l] \psi)(x) = \int (\zeta_l(y) - \zeta_l(x)) k_\tau(x - y, \epsilon) \psi(y) dy$$

and hence

$$|(\psi, \zeta_l [\mu_{x^\tau}(\epsilon), \zeta_l] \psi)| \leq \gamma \int |\zeta_l(x)| |\psi(x)| |h_\tau(x - y, \epsilon)| |\psi(y)| dx dy. \quad (2.32)$$

Secondly define

$$\zeta^l(x) = + \left(\beta^2 - \sum_{|l'| \leq l} \zeta_{l'}^2(x) \right)^{1/2}, \quad l \in \mathbb{Z}^+.$$

Obviously $1 \leq \zeta^l(x) \leq \beta$ and $\zeta^l \in O_M(\mathbb{R})$. We will show that ζ^l also satisfies an estimate of the form (2.32). Indeed we will show that

$$|\zeta^l(x) - \zeta^l(y)| \leq \alpha\gamma |x - y|.$$

Now

$$\zeta^l(x) - \zeta^l(y) = \left(\sum_{|l'| \leq l} \zeta_{l'}^2(y) - \zeta_{l'}^2(x) \right) (\zeta^l(x) + \zeta^l(y))^{-1}.$$

Since $\zeta^l(x) \geq 1$, this gives

$$\begin{aligned} |\zeta^l(x) - \zeta^l(y)| &\leq \frac{1}{2} \sup_{|l'| \leq l} |\zeta_{l'}(x) - \zeta_{l'}(y)| 2 \sup_x \sum_{|l'| \leq l} |\zeta_{l'}(x)| \\ &\leq \gamma\alpha |x - y|. \end{aligned}$$

Combining these estimates we obtain for arbitrary $L \in \mathbb{Z}^+$

$$\begin{aligned} &\sum_{|l| \leq L} (\psi, \zeta_l \mu_x^\tau(\epsilon) \zeta_l \psi) \\ &\leq \sum_{|l| \leq L} (\psi, \zeta_l \mu_x^\tau(\epsilon) \zeta_l \psi) + (\psi, \zeta^L \mu_x^\tau(\epsilon) \zeta^L \psi) \\ &\leq \sum_{|l| \leq L} (\psi, \mu_x^\tau(\epsilon) \zeta_l^2 \psi) + (\psi, \mu_x^\tau(\epsilon) (\zeta^L)^2 \psi) \\ &\quad + (\alpha\gamma + \alpha\beta\gamma) \int |\psi(x)| |\psi(y)| |h_\tau(x - y, \epsilon)| dx dy \\ &= \beta^2 (\psi, \mu_x^\tau(\epsilon) \psi) + (\alpha\gamma + \alpha\beta\gamma) \int |\psi(x)| |\psi(y)| |h_\tau(x - y, \epsilon)| dx dy. \end{aligned}$$

By Lemma 2.10, the Schwarz and the Young inequalities

$$\begin{aligned} \int |\psi(x)| |\psi(y)| |h_\tau(x - y, \epsilon)| dx dy &\leq \|\psi\|_2 \|\psi * h_\tau(\epsilon)\|_2 \\ &\leq \|\psi\|_2^2 \|h_\tau(\epsilon)\|_1. \end{aligned} \tag{2.34}$$

Combining (2.33) and (2.34) we obtain

$$\begin{aligned} \sum_{|l| \leq L} (\psi, \zeta_l \mu_x^\tau(\epsilon) \zeta_l \psi) &\leq \beta^2 (\psi, \mu_x^\tau(\epsilon) \psi) + \alpha\gamma(1 + \beta) \|h_\tau(\epsilon)\|_1 \|\psi\|_2^2 \\ &\leq \text{const}(\psi, \mu_x^\tau(\epsilon) \psi) \text{ uniformly in } L, \end{aligned}$$

proving the lemma.

If we consider $D(\mu_x^{\tau/2}(\epsilon))$ as a Hilbert space with the norm $\|\cdot\|_{\tau, \epsilon} = \|\mu_x^{\tau/2}(\epsilon) \cdot\|_2$,

then each ζ_l is a bounded map of $D(\mu_x^{\tau/2}(\epsilon))$ into itself and (2.29) may be rewritten as

$$\sum_{l \in \mathbb{Z}} \|\zeta_l \psi\|_{\tau, \epsilon}^2 \leq \text{const} \|\psi\|_{\tau, \epsilon}^2, \quad \psi \in D(\mu_x^{\tau/2}(\epsilon)), \quad (2.35)$$

since $\mathfrak{S}(\mathbb{R})$ is a core for the self-adjoint operator $\mu_x^{\tau/2}(\epsilon)$.

Lemma 2.11 and Theorem 3.2,1 in [8] give

PROPOSITION 2.12. *Let the conditions be as in Lemma 2.11. Then as an inequality between positive self-adjoint operators in Fock space*

$$\sum_{l \in \mathbb{Z}} N_{\tau, \zeta_l}^\epsilon \leq \text{const} N_\tau^\epsilon \leq \text{const} N_\tau \quad (0 \leq \tau < 1). \quad (2.36)$$

Here we have set $N_\tau = \sum_{\epsilon=0, \pm 1} N_\tau^\epsilon$ such that $N_\tau^\epsilon \leq N_\tau$ ($\epsilon = 0, \pm 1$).

3. LOCALIZED WICK MONOMIALS

In order to prove the volume-dependent first-order estimate, we will improve the standard proof of estimate (1.3) as done in [11] to give (1.6). This means we will use the method of approximative diagonalization of $H(g_n)$ and then give a new volume (i.e., n)-dependent estimate of the error terms occurring. Now these error terms are typically of the following form:

$$\begin{aligned} V(n) &= \int \tilde{g}_n \left(\sum_{i=1}^{r+r'} k_i \right) \prod_{i=1}^r b(x_i, \epsilon_i)^* \tilde{v}(k_1 \cdots k_r, k_{r+1} \cdots k_{r+r'}) \\ &\quad \times \prod_{i=r+1}^{r+r'} b(-k_i, \epsilon_i) dk_1 \cdots dk_{r+r'}, \end{aligned} \quad (3.1a)$$

$$\begin{aligned} V'(n) &= \int \tilde{g}_n \left(\sum_{i=1}^{r+r'} k_i + \sum_{i=1}^s q_i \right) \tilde{g}_n \left(\sum_{i=1}^{t+t'} \bar{k}_i + \sum_{i=1}^s q_i \right) \prod_{i=1}^r b(k_i, \epsilon_i)^* \prod_{i=1}^t b(\bar{k}_i, \epsilon_i)^* \\ &\quad \times \tilde{v}_1(k_1 \cdots k_{r+r'}; q_1 \cdots q_s) \tilde{v}_2(\bar{k}_1 \cdots \bar{k}_{t+t'}; q_1 \cdots q_s) \prod_{i=r+1}^{r+r'} b(-k_i, \epsilon_i) \\ &\quad \times \prod_{i=t}^{t+t'} b(-\bar{k}_i, \epsilon_i) \prod_{i=1}^{r+r'} dk_i \prod_{i=1}^{t+t'} d\bar{k}_i \prod_{i=1}^s dq_i. \end{aligned} \quad (3.1b)$$

Here

$$\tilde{g}_n(k) = \sum_{|j| \leq n} e^{ijk} \tilde{h}(k); \quad \tilde{h}(k) = (2\pi)^{-1/2} \int e^{-ikx} h(x) dx$$

and $\tilde{v}$, $\tilde{v}_1$, $\tilde{v}_2$ are such elements in $\mathfrak{S}'(\mathbb{R}^m)$ ($m = r + r'$, resp. $r + r' + s$, resp. $t + t' + s$), that $V(n)$ and $V'(n)$ are defined as operators on D .

Let us look at $V(n)$ in the x -space representation

$$V(n) = \int g_n(x) \prod_{i=1}^r b(x_i, \epsilon_i)^* v(x_1 - x, \dots, x_{r+r'} - x) \prod_{i=r+1}^{r+r'} b(x_i, \epsilon_i) dx dx_1 \dots dx_{r+r'} \quad (3.1c)$$

v being the Fourier transform of $\tilde{v}$.

Assuming $v(x_1 - x, \dots, x_{r+r'} - x)$ to be “small” if

$$\sum_{i=1}^{r+r'} |x_i - x|$$

is large, then with the decomposition

$$V(n) = \sum_{|j| \leq n} V_j \quad (3.2)$$

defined by the decomposition (1.14)

$$g_n = \sum_{|j| \leq n} h_j,$$

this means that the contribution to V_j from regions $(x_1 \dots x_{r+r'}) \in \mathbb{R}^{r+r'}$ with large $\sum_{i=1}^{r+r'} |x_i - j|$ will be “small.”

A similar property is expected for $V'(n)$, i.e., if we write

$$V'(n) = \sum_{\substack{|j| \leq n \\ |j'| \leq n}} V_{j,j'}$$

in the x -space representation with x_i being the canonically conjugate variables to k_i and $\bar{x}_i$ the ones to $\bar{k}_i$, then the contributions to $V_{j,j'}$ from regions of integration, where

$$\sum_{i=1}^{r+r'} |x_i - j| + \sum_{i=1}^{t+t'} |\bar{x}_i - j'|$$

is large, are expected to be “small.”

To make these ideas precise we will introduce the localization indices, which were used in a similar context for the $(\varphi^4)_2$ theory by J. Glimm and A. Jaffe [8]. Also the “smallness” will be defined in such a way that we may apply the local N_τ estimates in Chapter 2.

Let us start with an element $b \in C^\infty(\mathbb{R}^m)$ and let $D^{\alpha(m)}b$ be any partial derivative ($\alpha(m) = (\alpha_1, \dots, \alpha_m)$). Furthermore define

$$\tau(m) = (\tau_1, \dots, \tau_m) \quad (0 \leq \tau_i < 1)$$

and put

$$E_{\tau(m)}(\epsilon, \mathbf{k}) = \prod_{i=1}^m \mu(k_i, \epsilon_i)^{\tau_i/2}, \quad \mathbf{k} = (k_1, \dots, k_m). \quad (3.3)$$

Assume for given $\tau(m)$

$$E_{\tau(m)}^{-1}(\epsilon) D^{\alpha(m)}b \in L^2(\mathbb{R}^m) \quad \text{for all } \alpha(m). \quad (3.4)$$

Then in particular $b \in O_M(\mathbb{R}^m)$.

Formula (3.2) now suggests the following definition: The translate

$$b_j \in O_M(\mathbb{R}^m) (j \in Z) \text{ of } b$$

is defined to be

$$b_j(\mathbf{k}) = \exp \left(ij \sum_{i=1}^m k_i \right) b(\mathbf{k}). \quad (3.5)$$

Now we will localize b_j in each of its variables in x -space. For this purpose, let $\eta \in C_0^\infty(\mathbb{R})$ have the following properties:

- (a) $0 \leq \eta \leq 1$;
- (b) $\sum_{l \in Z} \eta(x - l) = 1 \quad \text{for all } x \in \mathbb{R}.$

Set $\eta_l(x) = \eta(x - l)$.

By condition (b), the η_l form a partition of unity. It is easy to construct an η having these properties. From now on η will be fixed. With help of the η_l we localize b_j in the following way: The Fourier transform of b_j is multiplied in each variable x_i (the conjugate variable to k_i) by the function η_{l_i} (l_i arbitrary). In k -space let $b_{j;l(m)}$ ($l(m) = (l_1, \dots, l_m)$) be the new element $\in O_M(\mathbb{R}^m)$ thus obtained. We call $l(m)$ a localization index. Since the η_l form a partition of unity, we have

$$\sum_{l(m)} b_{j;l(m)} = b_j \quad (3.7)$$

in $O_M(\mathbb{R}^m)$. We have explicitly

$$b_{j;l(m)}(\mathbf{k}) = \int \exp \left[ij \sum_{i=1}^m (k_i - k'_i) \right] b(\mathbf{k} - \mathbf{k}') \prod_{i=1}^m e^{ik'_i l_i} \tilde{\eta}(k'_i) dk'_i. \quad (3.8)$$

Therefore

$$\begin{aligned} \hat{b}_{j;l(m)}(\mathbf{k}) &\equiv \exp \left(-ij \sum_{i=1}^m k_i \right) b_{j;l(m)}(\mathbf{k}) \\ &= \int \exp \left[i \sum_{i=1}^m (l_i - j) k_i' \right] b(\mathbf{k} - \mathbf{k}') \prod_{i=1}^m \tilde{\eta}(k_i') dk_i' \end{aligned} \quad (3.9)$$

We expect $\hat{b}_{j;l(m)}$ to be small when $\sum_{i=1}^m |l_i - j|$ is large, so set

$$(l(m) - j)^{\alpha(m)} = \prod_{i=1}^m (l_i - j)^{\alpha_i}; \quad |l(m) - j| = \sum_{i=1}^m |l_i - j|,$$

$$|\alpha(m)| = \sum_{i=1}^m \alpha_i; \quad (\alpha - \beta)(m) = (\alpha_1 - \beta_1, \dots, \alpha_m - \beta_m), \quad \binom{\alpha(m)}{\beta(m)} = \prod_{i=1}^m \binom{\alpha_i}{\beta_i}.$$

Then

$$\begin{aligned} & (l(m) - j)^{\alpha(m)} (-i)^{|\alpha(m)|} \hat{b}_{j;l(m)}(\mathbf{k}) \\ &= \sum_{\beta(m)} \binom{\alpha(m)}{\beta(m)} \int \exp \left[i \sum_{i=1}^m (l_i - j) k_i' \right] D'^{\beta(m)} b(\mathbf{k} - \mathbf{k}') D'^{(\alpha-\beta)(m)} \prod_{i=1}^m \tilde{\eta}(k_i') dk_i', \end{aligned} \quad (3.10)$$

where D' denotes differentiation with respect to the variables $k_1', \dots, k_m'$.

If we now use the inequality

$$E_{\tau(m)}^{-1}(\epsilon, \mathbf{k}) \leq C(\tau(m)) E_{\tau(m)}^{-1}(\epsilon, \mathbf{k} - \mathbf{k}') E_{\tau(m)}(\epsilon, \mathbf{k}') \quad (3.11)$$

which just states that $E_{\tau(m)}^{-1}(\epsilon, \mathbf{k})$ is a temperate weight function in the sense of L. Hörmander [13], and Young's inequality

$$\|f * f'\|_2 \leq \|f\|_1 \|f'\|_2, \quad (3.12)$$

then Eq. (3.10) gives the following

LEMMA 3.1. *Assume $b \in C^\infty(\mathbb{R}^m)$ such that for some $\tau(m)$ and all $\alpha(m) E_{\tau(m)}^{-1}(\epsilon) D^{\alpha(m)} b \in L^2(\mathbb{R}^m)$. Then $E_{\tau(m)}^{-1}(\epsilon) D^{\alpha(m)} \hat{b}_{j;l(m)} \in L^2(\mathbb{R}^m)$ for all $l(m)$, $\alpha(m)$, j and the same $\tau(m)$ such that*

$$\begin{aligned} \|E_{\tau(m)}^{-1}(\epsilon) D^{\alpha(m)} \hat{b}_{j;l(m)}\|_2 &\leq C(\alpha(m), \beta)(|l(m) - j| + 1)^{-\beta} \\ &\quad \times \sup_{\gamma(m): |\gamma(m)| \leq |\alpha(m)| + \beta} \|E_{\tau(m)}^{-1}(\epsilon) D^{\gamma(m)} b\|_2 \quad \text{for all } \beta. \end{aligned} \quad (3.13)$$

Lemma 3.1 will help us estimate operators of the form (3.1a). Note that for $\alpha(m) = 0$, we may substitute $b_{j;l(m)}$ in (3.13). This gives the estimate for localized kernels exactly of the form needed in order to apply the local N_τ estimates of Chapter 2.

We will also need the following

Remark 3.2. If we suspend the condition $\tau_i \leq 1$, the proof of Lemma 3.1 implies in particular the following: To each $\mathfrak{S}(\mathbb{R}^m)$ -norm s and each $\beta \in Z^+$ there exists another $\mathfrak{S}(\mathbb{R}^m)$ -norm $s' = s'(s, \beta)$ such that

$$\| \hat{b}_{j;l(m)} \|_s \leq (|l(m) - j| + 1)^{-\beta} \| b \|_{s'} . \quad (3.14)$$

To estimate operators of the form (3.1b), we need another lemma, relying on the same ideas which led to Lemma 3.1.

Let $b^{(\nu)} \in C^\infty(\mathbb{R}^{m_\nu+s})$ ($\nu = 1, 2; s \geq 1, m_\nu \geq 0$) be such that

$$E_{\tau^{(\nu)}(m_\nu)}^{-1}(\epsilon^{(\nu)}) D^{\alpha(m_\nu+s)} b^{(\nu)} \in L^2(\mathbb{R}^{m_\nu+s})$$

for some $\tau^{(\nu)}(m_\nu)$ and all $\alpha(m_\nu + s)$.

Define

$$(b_j^{(1)} \Delta_s b_{j'}^{(2)})(k_1 \cdots k_{m_1+m_2}) \\ = \int b_j^{(1)}(k_1 \cdots k_{m_1}; q_1 \cdots q_s) b_{j'}^{(2)}(k_{m_1+1}, \cdots k_{m_1+m_2}; q_1 \cdots q_s) dq_1 \cdots dq_s .$$

Setting

$$\tau^{(1,2)}(m_1 + m_2) = (\tau_1^{(1)} \cdots, \tau_{m_1}^{(1)} \cdots \tau_{m_2}^{(2)}); \quad \epsilon^{(1,2)} = (\epsilon_1^{(1)}, \cdots \epsilon_{m_1}^{(1)}, \epsilon_1^{(2)} \cdots \epsilon_{m_2}^{(2)})$$

we have in particular

$$E_{\tau^{(1,2)}(m_1+m_2)}^{-1}(\epsilon^{(1,2)})(b_j^{(1)} \Delta_s b_{j'}^{(2)}) \in L^2(\mathbb{R}^{m_1+m_2}) \quad \text{for all } j, j'.$$

Furthermore write

$$l^{(1,2)}(m_1 + m_2) = (l_1^{(1)}, \cdots l_{m_1}^{(1)}, l_1^{(2)} \cdots, l_{m_2}^{(2)}) \in Z^{(m_1+m_2)}$$

$$\text{for } l^{(\nu)}(m_\nu) \in Z^{m_\nu} \quad (\nu = 1, 2)$$

and set

$$(b_j^{(1)} \Delta_s b_{j'}^{(2)})_{l^{(1,2)}(m_1+m_2)}(k_1, \cdots k_{m_1+m_2}) \\ = \exp \left(-ij \sum_{i=1}^{m_1} k_i - ij' \sum_{i=m_1+1}^{m_1+m_2} k_i \right) (b_j^{(1)} \Delta_s b_{j'}^{(2)})_{l^{(1,2)}(m_1+m_2)}(k_1, \cdots k_{m_1+m_2}).$$

Then we have the

LEMMA 3.3. Assume $b^{(\nu)} \in C^\alpha(\mathbb{R}^{m_\nu+s})(\nu = 1, 2; s \geq 1, m_\nu \geq 0)$ is such that for some $\tau^{(\nu)}(m_\nu)$,

$$E_{\tau^{(\nu)}(m_\nu)}^{-1}(\epsilon^{(\nu)}) D^{\alpha(m_\nu+s)} b^{(\nu)} \in L^2(\mathbb{R}^{m_\nu+s})$$

for all $\alpha(m_\nu + s)$. Then

$$E_{\tau^{(1,2)}(m_1+m_2)}^{-1}(\epsilon^{(1,2)}) D^{\alpha(m_1+m_2)} (b_j^{(1)} \Delta_s b_{j'}^{(2)})_{l^{(1,2)}(m_1+m_2)} \in L^2(\mathbb{R}^{m_1+m_2})$$

for all $j, j', \alpha(m_1 + m_2), l^{(\nu)}(m_\nu) \in Z^{m_\nu}(\nu = 1, 2)$ such that the following estimate is true

$$\begin{aligned} & \| E_{\tau^{(1,2)}(m_1+m_2)}^{-1}(\epsilon^{(1,2)}) D^{\alpha(m_1+m_2)} (b_j^{(1)} \Delta_s b_{j'}^{(2)})_{l^{(1,2)}(m_1+m_2)} \|_2 \\ & \leq C(\alpha(m_1 + m_2), \beta_1, \beta_2, \beta_3)(|l^{(1)}(m_1) - j| + 1)^{-\beta_1} \\ & \quad \times (|l^{(2)}(m_2) - j'| + 1)^{-\beta_2} (|j + j'| + 1)^{-\beta_3} \\ & \quad \times \sup' \| E_{\tau^{(1)}(m_1)}^{-1}(\epsilon^{(1)}) D^{\gamma^{(1)}(m_1+s)} b^{(1)} \|_2 \| E_{\tau^{(2)}(m_2)}^{-1}(\epsilon^{(2)}) D^{\gamma^{(2)}(m_2+s)} b^{(2)} \|_2, \end{aligned} \quad (3.15)$$

where $\sup'$ is the supremum over all $\gamma^{(1)}(m_1 + s), \gamma^{(2)}(m_2 + s)$ with

$$|\gamma^{(1)}(m_1 + s)| + |\gamma^{(2)}(m_2 + s)| \leq |\alpha(m_1 + m_2)| + \beta_1 + \beta_2 + \beta_3$$

for given $\beta_i \in Z^+$ and $\alpha(m_1 + m_2)$

4. THE VOLUME-DEPENDENT FIRST-ORDER N_τ ESTIMATE

In order to prove the estimate (1.6) with $g = g_n$ it is of course sufficient to prove a similar estimate for the cut-off Hamiltonian $H(g_n, \kappa)$ which is uniform in κ . More precisely κ will be introduced in the following way. Let χ be a nonnegative, symmetric function $\epsilon C_0^\infty(\mathbb{R})$, such that $\chi(0) = 1$. Then a cut off in momentum space with parameter $\kappa(\kappa > 0)$ of a kernel $b \in O_M(\mathbb{R}^m)$ is obtained by multiplication with $\chi_\kappa(\mathbf{k}) = \prod_{i=1}^m \chi(k_i/\kappa)$

$$b_\kappa(\mathbf{k}) = b(\mathbf{k}) \chi_\kappa(\mathbf{k}). \quad (4.1)$$

(We hope this will not be confused with the translate b_i of b .) This gives an admissible ultraviolet cut-off in the sense of Definition 6.12 in [9].

Since we want to obtain uniform estimates in κ of the error terms arising from

the partial diagonalization of $H(g_n, \kappa)$ by means of the results of the previous chapters, the following lemma will be useful:

LEMMA 4.1. Assume $b \in C^\infty(\mathbb{R}^m)$ such that $E_{\tau(m)}^{-1}(\epsilon) D^{\alpha(m)} b \in L^2(\mathbb{R}^m)$ for some $\tau(m)$ and all $\alpha(m)$. Then also $E_{\tau(m)}^{-1}(\epsilon) D^{\alpha(m)} b_\kappa \in L^2(\mathbb{R}^m)$ for the same $\tau(m)$, all $\alpha(m)$ and κ . The following κ -independent estimates are true:

$$\| E_{\tau(m)}^{-1}(\epsilon) D^{\alpha(m)} b_\kappa \|_2 \leq c(\alpha(m)) \sup_{\beta(m): |\beta(m)| \leq |\alpha(m)|} \| E_{\tau(m)}^{-1}(\epsilon) D^{\beta(m)} b \|_2. \quad (4.2)$$

Proof. This is immediate from Leibniz' rule, since

$$\| D^{\nu(m)} \chi_\kappa \|_\infty \leq c'(\gamma(m))(1 + \kappa)^{-|\nu(m)|}$$

uniformly in κ .

In the course of the proof of (1.6), we will also introduce a lower cut-off parameter ρ ($\rho > -1$). Let Θ_ρ be the following function: Choose first Θ to be a positive monotonically decreasing function $\epsilon C^\infty(-\epsilon, +1 + \epsilon)$ ($\epsilon > 0$) such that $\Theta(x) = 1$ for $x \leq 0$ and $\Theta(x) = 0$ for $x \geq 1$.

For $\rho \geq 0$ set

$$\Theta_\rho(k) = \begin{cases} 1 & \text{if } |k| \leq \rho \\ \Theta(|k| - \rho) & \text{if } \rho \leq |k| \leq \rho + 1 \\ 0 & \text{if } |k| \geq \rho + 1 \end{cases}$$

and for $-1 \leq \rho \leq 0$

$$\Theta_\rho(k) = \Theta(|k| - \rho).$$

Of course Θ_ρ is an approximation to the characteristic function of the interval $[-\rho, +\rho]$ if $\rho > 0$. The essential estimate connected with the lower cut off ρ is

LEMMA 4.2. Let $b \in C^\infty(\mathbb{R}^m)$ be such that $E_{\tau(m)}^{-1}(\epsilon) D^{\alpha(m)} b \in L^2(\mathbb{R}^m)$ for some $\tau(m)$ and all $\alpha(m)$. Set $(1 - \Theta_\rho) b(\mathbf{k}) = (1 - \Theta_\rho)(k_1) b(\mathbf{k})$. Then for any $\tau^1(m)$, with $1 \geq \tau_i^1 \geq \tau_i$ for all i , and for all all $n \in \mathbb{Z}^+$ there exists $c(n)$ such that

$$\| E_{\tau^1(m)}^{-1}(\epsilon) D^{\alpha(m)} (1 - \Theta_\rho) b \|_2 \leq c(n) (1 + \rho)^{-\frac{1}{2}(\tau_1' - \tau_1)} \quad (4.3)$$

for all $\alpha(m)$ with $|\alpha(m)| \leq n$.

The proof is trivial. Closely related is the following lemma, which will be used together with Lemma 2.7 and Remark 3.2.

LEMMA 4.3. Let the hypothesis be as in lemma 4.2. Then for given $\delta > 0$ and $n \in \mathbb{Z}^+$ there exists $b' \in \mathfrak{S}(\mathbb{R}^m)$ such that with the same $\tau(m)$

$$\| E_{\tau(m)}^{-1}(\epsilon) D^{\alpha(m)} (b_\kappa - b'_\kappa) \|_2 \leq \delta \quad (4.4)$$

for all $\alpha(m)$ with $|\alpha(m)| \leq n$ uniformly in κ . Also there exist $c(s) < \infty$ (for given b) such that

$$\|b_{\kappa}'\|_s \leq c(s) \quad (4.5)$$

uniformly in κ for any $\mathfrak{S}(\mathbb{R}^m)$ -norm s .

Proof. It is sufficient to choose $b'(k) = \prod_{i=1}^m \theta_{\rho}(k_i) b(k)$ with ρ sufficiently large.

We turn to the explicit definition and discussion of $H(g_n, \kappa)$. From now on k will denote a boson momentum, p a fermion momentum and we set

$$\begin{aligned} \mu(k) &= \mu(k, 0); & \omega(p) &= \mu(p, \pm 1) \text{ or for short } \mu = \mu(k), \\ \omega_i &= \omega(p_i), \\ a(k) &= b(k, 0); & b(p) &= b(p, +1), b'(p) = b(p, -1). \end{aligned}$$

First we write [see (1.2)]

$$H_I(g_n) = W^C(n) + W(n) + W^A(n), \quad (4.6)$$

where the pair-creation terms $W^C(n)$ equal

$$W^C(n) = \sum_{|j| \leq n} \int \{a(k)^* + a(-k)\} b(p_1)^* b'(p_1)^* w_j^C(k, p_1, p_2) dk dp_1 dp_2$$

and the kernel w^C equals

$$\begin{aligned} w^C(k, p_1, p_2) &= -\frac{\lambda}{4\pi} \tilde{h}(k + p_1 + p_2)(\mu\omega_1\omega_2)^{-1/2} (\omega_1\omega_2 - p_1p_2 - m_f^2)^{1/2} \text{sign}(p_1 - p_2) \\ &= -w^C(-k_1, -p_1, -p_2) = \bar{w}^C(k, p_1, p_2). \end{aligned}$$

The pair-annihilation terms are $W^A(n) = W^C(n)^*$,

$$W^A(n) = \sum_{|j| \leq n} \int w_j^C(k, p_1, p_2) b(-p_1) b'(-p_2) \{a(k)^* + a(-k)\} dk dp_1 dp_2.$$

The boson emission and absorption terms $W(n)$ equal

$$\begin{aligned} W(n) &= \sum_{|j| \leq n} w_j(k, p_1, p_2) \{a(k)^* + a(-k)\} \\ &\quad \times \{b(p_1)^* b(-p_2) + b'(p_1)^* b'(-p_2)\} dk dp_1 dp_2, \end{aligned}$$

where

$$w(k, p_1, p_2) = \frac{-\lambda}{4\pi} \tilde{h}(k + p_1 + p_2)(\mu\omega_1\omega_2)^{-1/2} (\omega_1\omega_2 + p_1p_2 + m_f^2)^{1/2}.$$

w_j^C and w_j are the translates of w^C and w as defined in Chapter 3.

Now let $w_{\kappa,\rho}^C$ and $w_{\kappa,\rho}$ be defined by

$$w_{\kappa,\rho}^C(k, p_1, p_2) = w^C(k, p_1, p_2) \chi_\kappa(k, p_1, p_2)(1 - \Theta_\rho(p_1 - p_2)),$$

$$w_{\kappa,\rho}(k, p_1, p_2) = w(k, p_1, p_2) \chi_\kappa(k, p_1, p_2)(1 - \Theta_\rho(p_1 - p_2)),$$

and write $H_I(g_n, \kappa, \rho)$ for the expression obtained from (4.6) by the substitution

$$w^C \rightarrow w_{\kappa,\rho}^C \quad \text{and} \quad w \rightarrow w_{\kappa,\rho}.$$

Finally, we introduce the counterterm by

$$\mathfrak{c}(g_n, \kappa, \rho) = -\frac{1}{2}\delta m^2(\kappa, \rho) \int :\varphi(x)^2: g_n(x)^2 dx - E(g_n, \kappa, \rho). \quad (4.7)$$

Here

$$\delta m^2(\kappa, \rho) = \frac{-\lambda}{2\pi} \int \left| \chi_\kappa\left(0, \frac{\xi}{2}, -\frac{\xi}{2}\right) \right|^2 \omega(\xi)^{-1} (1 - \Theta_\rho(\xi))^2 d\xi \quad (4.8)$$

is independent of the spacial cut off g_n . Also

$$\begin{aligned} E(g_n, \kappa, \rho) &= \sum_{|j|, |j'| \leq n} \int |w_{\kappa,\rho}^C(k, p_1, p_2)|^2 e^{i(j+j')(k+p_1+p_2)} (\mu + \omega_1 + \omega_2)^{-1} dk dp_1 dp_2. \end{aligned} \quad (4.9)$$

We define the renormalized cut-off Hamiltonian as

$$H(g_n, \kappa, \rho) = H_0 + H_I(g_n, \kappa, \rho) + \mathfrak{c}(g_n, \kappa, \rho) \quad (4.10)$$

and

$$H(g_n, \kappa) = H(g_n, \kappa, -1) \quad (4.11)$$

and correspondingly $\mathfrak{c}(g_n, \kappa)$ etc.

Now define

$$\begin{aligned} \Gamma W_{\kappa,\rho}^C(n) &= \sum_{|j| \leq n} \int w_{\kappa,\rho}^C(k, p_1, p_2) \{a(k)^* (\mu + \omega_1 + \omega_2)^{-1} \\ &\quad + a(k)(\omega_1 + \omega_2)^{-1}\} b(p_1)^* b'(p_2)^* dk dp_1 dp_2 \end{aligned}$$

and

$$\dot{b}(k, \epsilon) = b(k, \epsilon) + [b(k, \epsilon), \Gamma W_{\kappa,\rho}^C(n)].$$

Let $1 - \tau$ and c be sufficiently small so that for all k

$$0 \leq \bar{\mu}(k, \epsilon) = \mu(k, \epsilon) - c\mu(k, \epsilon)^\tau; \epsilon = 0, \pm 1.$$

We define a positive approximation to $H(g_n, \kappa, \rho)$ by

$$H(n, \kappa, \rho) = \sum_{\epsilon=0, \pm 1} \int \hat{b}(k, \epsilon)^* \bar{\mu}(k, \epsilon) \hat{b}(k, \epsilon) dk. \quad (4.12)$$

For the motivation see [6] and [11].

Expanding the right-hand side of (4.12) in terms of Wick-ordered monomials, we obtain

$$0 \leq H(n, \kappa, \rho) = -cN_\tau + H_1(n, \kappa, \rho) + H_2(n, \kappa, \rho). \quad (4.13)$$

As in [11] (to which the reader is referred) we define $H_1(n, \kappa, \rho)$ to be the contribution which resembles $H(g_n, \kappa, \rho)$. We will prove

LEMMA 4.4. *For given $\epsilon > 0$ there exists $\rho_1 = \rho_1(\epsilon)$ and $c' = c'(\epsilon)$ such that uniformly in n and κ and for all $\rho > \rho_1(\epsilon)$*

$$\pm(H_1(n, \kappa, \rho) - H(g_n, \kappa, \rho)) \leq \epsilon N_\tau + c'n, \quad (4.14)$$

$$H_2(n, \kappa, \rho) \leq \epsilon N_\tau + c'n. \quad (4.15)$$

LEMMA 4.5. *Let $\epsilon > 0$ and choose ρ_1 as defined by Lemma 4.4. Then there exists $c'' = c''(\epsilon)$ such that uniformly in n, κ and $\rho \leq \rho_1$*

$$0 \leq \epsilon N_\tau + \{H(g_n, \kappa, \rho) - H(g_n, \kappa, \rho_1)\} + c''n. \quad (4.16)$$

The combination of Lemmas 4.4 and 4.5 of course proves the first-order volume-dependent N_τ estimate (1.6) for $g = g_n$, and hence in general by the remarks made in Chapter 1. Indeed we have uniformly in n and κ

$$0 \leq (c - 3\epsilon) N_\tau \leq H(g_n, \kappa) + (2c' + c'') \cdot n. \quad (4.17)$$

In order to demonstrate the method of proof, it is convenient to start with the proof of Lemma 4.5 first, assuming the existence of a fixed ρ_1 making Lemma 4.4 valid.

In the sequel c_i will be estimate constants independent of n and κ . First set

$$\delta H = H(g_n, \kappa, \rho) - H(g_n, \kappa, \rho_1)$$

then

$$\begin{aligned} \delta H = & \delta W^C + \delta W + \delta W^A - \frac{1}{2}(\delta m^2(\kappa, \rho) - \delta m^2(\kappa, \rho_1)) \int : \varphi(x)^2 : g_n(x)^2 dx \\ & - (E(g_n, \kappa, \rho) - E(g_n, \kappa, \rho_1)). \end{aligned}$$

We note that $0 \leq \Theta_\rho \leq \Theta_{\rho_1} \leq 1$ for all $\rho \leq \rho_1$. Hence

$$(1 - \Theta_\rho)^2(\xi) - (1 - \Theta_{\rho_1})^2(\xi) \geq 0$$

with equality for $|\xi| \geq \rho_1 + 1$, such that the defining equation (4.8) for δm^2 gives

$$0 \leq -\delta m^2(\kappa, \rho) + \delta m^2(\kappa, \rho_1) \leq c_1 \ln(\rho_1 + 1) \text{ uniformly in } \kappa \text{ and } \rho \leq \rho_1. \quad (4.18)$$

Now for given ρ_1 and $\epsilon' > 0$ the following estimate is known (see [8])

$$\int :\varphi(x)^2: g_n(x)^2 dx \geq -\epsilon'(c_1 \ln(\rho_1 + 1))^{-1} N - c_2 \cdot n. \quad (4.19)$$

Thus estimates (4.18) and (4.19) take care of the mass-renormalization contribution to δH . Equation (4.19) may also be proved with an argument used by J. Glimm [5, Section 3] and methods which will be elaborated below. Next we look at the contribution to δH arising from the energy-renormalization terms. Set

$$\xi = p_1 - p_2; \quad \eta = p_1 + p_2$$

$$\begin{aligned} & -E(g_n, \kappa, \rho) + E(g_n, \kappa, \rho_1) \\ &= \sum_{\substack{|j| \leq n \\ |j'| \leq n}} \int e^{i(j+j')(k+p_1+p_2)} |w^C(k, p_1, p_2)|^2 (\mu + \omega_1 + \omega_2)^{-1} \chi_n(k, p_1, p_2)^2 \\ & \quad \times ((1 - \Theta_\rho(\xi))^2 - (1 - \Theta_{\rho_1}(\xi))^2) dk dp_1 dp_2. \end{aligned}$$

Using the estimate

$$|(\omega_1 \omega_2) - \frac{1}{2}(\omega_1 \omega_2 - p_1 p_2 - m_f^2)^{1/2}| \leq c_3 \omega(\xi)^\delta \omega(\eta)^{-\delta/2} \quad (0 < \delta < 1) \quad (4.20a)$$

and Lemma 4.1 gives

$$\begin{aligned} & |E(g_n, \kappa, \rho) - E(g_n, \kappa, \rho_1)| \\ & \leq c_4 \sum_{\substack{|j| \leq n \\ |j'| \leq n}} (|j + j'| + 1)^{-2} \\ & \quad \times \sup_{0 \leq m \leq 2} \left\| \frac{\partial^m}{\partial k^m} ((1 - \Theta_\rho(\xi))^2 - (1 - \Theta_{\rho_1}(\xi))^2)^{1/2} (\mu + \omega_1 + \omega_2)^{-1/2} w^C \right\|_2^2 \\ & \leq c_5 \cdot n \text{ uniformly in } \kappa, n \text{ and } \rho \leq \rho_1. \end{aligned} \quad (4.21)$$

We turn to the discussion of δW^C , δW and δW^A . We will be rather detailed in the discussion of these terms. Of course it is sufficient to consider δW^C and δW , δW^A being the adjoint of δW^C . We will use the local N_τ estimates. In order to combine the results of Chapter 2 with those of Chapter 3, consider the function

$\eta \in C_0^\infty(\mathbb{R})$ which we used to introduce the localization indices and which had the properties (3.6). Fix $\zeta \in C_0^\infty(\mathbb{R}) (0 \leq \zeta \leq 1)$ such that $\zeta\eta = \eta$. Note that $\zeta_l\eta_l = \eta_l$ for all $l \in \mathbb{Z}$ if we set $\zeta_l(x) = \zeta(x - l)$ and that the conditions of Lemma 2.11 are trivially satisfied for the ζ_l .

Agreeing that the functions Θ_ρ and Θ_{ρ_1} always have $\xi = p_1 - p_2$ as argument, then for given $\epsilon'' > 0$ choose $\delta w^{C,1}, \delta w^1 \in \mathfrak{S}(\mathbb{R}^3)$ such that

$$\sup_{|\alpha(3)| \leq 8} \|\omega_1^{-\tau/2} D^{\alpha(3)} \{w^C(\Theta_{\rho_1} - \Theta_\rho) - \delta w^{C,1}\}\|_2 \leq \epsilon'' \quad (4.22a)$$

[due to estimate (4.20a), this is possible],

$$\sup_{|\alpha(3)| \leq 8} \|\omega_1^{-\tau/2} D^{\alpha(3)} \{w(\Theta_{\rho_1} - \Theta_\rho) - \delta w^1\}\|_2 \leq \epsilon''. \quad (4.22b)$$

[This is possible due to the estimate

$$|(\omega_1\omega_2)^{-1/2} (\omega_1\omega_2 + p_1p_2 + m_f^2)^{1/2}| \leq \text{const.}] \quad (4.20b)$$

Set

$$\delta w^{C,2} = w^C(\Theta_{\rho_1} - \Theta_\rho) - \delta w^{C,1};$$

$$dw_\kappa^{C,i}(k, p_1, p_2) = \delta w^{C,i}(k, p_1p_2) \chi_\kappa(k, p_1, p_2),$$

$$\delta w^2 = w(\Theta_{\rho_1} - \Theta_\rho) - \delta w^1;$$

$$\delta w_\kappa^i(k, p_1, p_2) = \delta w^i(k, p_1, p_2) \chi_\kappa(k, p_1, p_2) \quad (i = 1, 2)$$

such that

$$\sum_{|j| \leq n} \sum_{i=1}^2 \delta w_{\kappa,j}^{C,i} \left(\text{resp. } \sum_{|j| \leq n} \sum_{i=1}^2 \delta w_{\kappa,j}^i \right)$$

is the kernel of δW^C (resp. δW).

Let Φ be in D . Upon introduction of localization indices $l(3) = (l_1, l_2, l_3) \in \mathbb{Z}^3$ (l_1 for the boson, l_2, l_3 for the fermions), the decomposition (3.7) gives

$$\begin{aligned} |(\Phi, \delta W^C \Phi)| &\leq \sum_{i=1}^2 \sum_{|j| \leq n} \sum_{l(3) \in \mathbb{Z}^3} \left| \left(\Phi, \int \delta w_{\kappa,j;l(3)}^{C,i}(k, p_1, p_2) (a(k)^* + a(-k)) \right. \right. \\ &\quad \left. \left. \times b(p_1)^* b'(p_2)^* dk dp_1 dp_2 \Phi \right) \right|, \end{aligned}$$

$$\begin{aligned} |(\Phi, \delta W \Phi)| &\leq \sum_{i=1}^2 \sum_{|j| \leq n} \sum_{l(3) \in \mathbb{Z}^3} \left| \left(\Phi, \int \delta w_{\kappa,j;l(3)}^i(k, p_1, p_2) (a(k)^* + a(-k)) \right. \right. \\ &\quad \left. \left. \times (b(p_1)^* b(-p_2) + b'(p_1)^* b'(-p_2)) dk dp_1 dp_2 \Phi \right) \right|. \end{aligned}$$

Let us write the right-hand sides as $\sum_{i=1}^2 A_i^C(\Phi, \Phi)$ (resp. $\sum_{i=1}^2 A_i(\Phi, \Phi)$). Then we have by Lemma 2.7 and the remarks 2.8 and 3.2

$$\begin{aligned} A_1^C(\Phi, \Phi) + A_1(\Phi, \Phi) &\leq c_6 \|\Phi\| \sum_{|j| \leq n} \sum_{l(3) \in \mathbb{Z}^3} \|(N_{0,l_1}^0 + 1)^{1/2} \Phi\| (|l_1 - j| + 1)^{-2} \\ &\quad \times (|l_2 - j| + 1)^{-2} (|l_3 - j| + 1)^{-2}, \end{aligned} \quad (4.23)$$

where the κ -independence of c_6 is guaranteed by Lemma 4.3 [see Eq. (4.5)]. Hence

$$\begin{aligned} A_1^C(\Phi, \Phi) + A_1(\Phi, \Phi) &\leq c_7 \|\Phi\| \sum_{|j| \leq n} \sum_{l_1 \in \mathbb{Z}} (\|(N_{0,l_1}^0)^{1/2} \Phi\| + \|\Phi\|) (|l_1 - j| + 1)^{-2} \\ &\leq c_7 \|\Phi\| (c_8 \cdot n^{1/2} \|N^{1/2} \Phi\| + c_9 n \|\Phi\|) \\ &\leq \epsilon' \|N^{1/2} \Phi\|^2 + (\epsilon')^{-1} c_{10} \cdot n \|\Phi\| \quad \text{for arbitrary } \epsilon' > 0 \end{aligned} \quad (4.24)$$

Here we used $\|(N_{0,l_1}^0 + 1)^{1/2} \Phi\| \leq \|(N_{0,l_1}^0)^{1/2} \Phi\| + \|\Phi\|$. Also by Schwarz inequality and Proposition 2.12

$$\begin{aligned} &\sum_{\substack{l_1 \\ |j| \leq n}} \|(N_{0,l_1}^0)^{1/2} \Phi\| (|l_1 - j| + 1)^{-2} \\ &\leq \left(\sum_{\substack{l_1 \\ |j| \leq n}} \|(N_{0,l_1}^0)^{1/2} \Phi\|^2 (|l_1 - j| + 1)^{-2} \right)^{1/2} \left(\sum_{\substack{l_1 \\ |j| \leq n}} (|l_1 - j| + 1)^{-2} \right)^{1/2} \\ &\leq c_8 \|N^{1/2} \Phi\| n^{1/2}. \end{aligned}$$

We turn to $A_2^C(\Phi, \Phi)$ and $A_2(\Phi, \Phi)$. By Lemma 2.5, Remark 2.6, Lemma 3.1 and the estimates (4.22a and b)

$$\begin{aligned} A_2^C(\Phi, \Phi) + A_2(\Phi, \Phi) &\leq \epsilon'' c_{11} \sum_{\substack{|j| \leq n \\ l(3) \in \mathbb{Z}^3}} \|(N_{\tau,l_2}^+ + 1)^{1/2} \Phi\| \|(N_{0,l_1}^0 + 1)^{1/2} \Phi\| \\ &\quad \times (|l_1 - j| + 1)^{-4} (|l_2 - j| + 1)^{-2} (|l_3 - j| + 1)^{-2}. \end{aligned}$$

The κ -independence of c_{11} is guaranteed by Lemma 4.1. First we have

$$\begin{aligned} A_2^C(\Phi, \Phi) + A_2(\Phi, \Phi) &\leq \epsilon'' c_{11} \sum_{\substack{|j| \leq n \\ l(3) \in \mathbb{Z}^3}} (\|(N_{\tau,l_2}^+)^{1/2} \Phi\| + \|\Phi\|) (\|(N_{0,l_1}^0)^{1/2} \Phi\| + \|\Phi\|) \\ &\quad \times (|l_1 - j| + 1)^{-4} (|l_2 - j| + 1)^{-2} (|l_3 - j| + 1)^{-2}. \end{aligned}$$

Hence estimating similarly as in (4.24)

$$\begin{aligned} A_2^C(\Phi, \Phi) + A_2(\Phi, \Phi) \\ \leq \epsilon'' c_{12} (\|N_\tau^{1/2} \Phi\|^2 + c_{13} n \|\Phi\|^2 \\ + c_{14} \sum_{l_1, l_2} (|l_1 - l_2| + 1)^{-2} \|(N_{0, l_1}^0)^{1/2} \Phi\| \|(N_{\tau, l_2}^+)^{1/2} \Phi\|). \end{aligned}$$

Now by the Schwarz inequality and Proposition 2.12

$$\begin{aligned} \sum_{l_1, l_2} (|l_1 - l_2| + 1)^{-2} \|(N_{0, l_1}^0)^{1/2} \Phi\| \|(N_{\tau, l_2}^+)^{1/2} \Phi\| \\ \leq \left(\sum_{l_1, l_2} (|l_1 - l_2| + 1)^{-2} \|(N_{0, l_1}^0)^{1/2} \Phi\|^2 \right)^{1/2} \\ \times \left(\sum_{l_1, l_2} (|l_1 - l_2| + 1)^{-2} \|(N_{\tau, l_2}^+)^{1/2} \Phi\|^2 \right)^{1/2} \\ \leq c_{15} \|N^{1/2} \Phi\| \|N_\tau^{1/2} \Phi\| \leq c_{16} \|N_\tau^{1/2} \Phi\|^2, \end{aligned}$$

so finally for arbitrary $\epsilon' > 0$

$$\begin{aligned} A_2^C(\Phi, \Phi) + A_2(\Phi, \Phi) &\leq \epsilon'' c_{17} \|N_\tau^{1/2} \Phi\|^2 + c_{18} \cdot n \|\Phi\|^2 \\ &\leq \epsilon' \|N_\tau^{1/2} \Phi\|^2 + c_{18} \cdot n \|\Phi\|^2 \end{aligned} \quad (4.25)$$

if we choose $\epsilon'' = \epsilon'(c_{17})^{-1}$.

Since ϵ' was arbitrary, the combination of Estimates (4.23)–(4.25) completes the proof of Lemma 4.5 if we choose $5\epsilon' = \epsilon$. We turn to the

Proof of Lemma 4.4.

Let us first look at $(H_1 - H)(g_n, \kappa, \rho)$. With the obvious notation

$$H_I(g_n, \kappa, \rho) = W^C(g_n, \kappa, \rho) + W(g_n, \kappa, \rho) + W^H(g_n, \kappa, \rho).$$

We have (see [11, Lemma 7.1.2])

$$(H_1 - H)(g_n, \kappa, \rho) = -W(g_n, \kappa, \rho) + \delta M.$$

Here δM will give the mass-renormalization cancellation, which we will write down and discuss in a moment. By construction the vacuum-energy renormalization cancels exactly. Now

$$\|(\mu\omega_1)^{-\tau'/2} D^{\alpha(3)}_{\mathcal{W}}\|_2 + \|(\mu\omega_2)^{-\tau'/2} D^{\alpha(3)}_{\mathcal{W}}\|_2 \leq C(\alpha(3)) < \infty$$

for τ' close to one. Since we also have chosen τ to be close to one, we may assume $\tau' = \tau - \delta$ ($\delta > 0$) and hence

$$\sup_{|\alpha(3)| \leq 8} \|(\mu\omega_1)^{-\tau/2} D^{\alpha(3)} w(1 - \Theta_\rho)\|_2 + \sup_{|\alpha(3)| \leq 8} \|(\mu\omega_2)^{-\tau/2} D^{\alpha(3)} w(1 - \Theta_\rho)\|_2 \\ \leq c_{19}(\rho + 1)^{-\delta/2}$$

by Lemma 4.2. Thus Lemmas 2.4 and 3.1 give

$$|(\Phi, W(g_n, \kappa, \rho) \Phi)| \leq c_{20}(\rho + 1)^{-\delta/2} \sum_{\substack{|j| \leq n \\ l(3) \in \mathbb{Z}^3}} \|(N_{\tau, l_1}^0 + 1)^{1/2} \Phi\| \\ \times (\|(N_{\tau, l_2}^+ + 1)^{1/2} \Phi\| + \|(N_{\tau, l_2}^- + 1)^{1/2} \Phi\|) \\ \times (|l_1 - j| + 1)^{-4} (|l_2 - j| + 1)^{-2} (|l_3 - j| + 1)^{-2}$$

for $\Phi \in D$. Again c_{20} may be chosen to be κ -independent by Lemma 4.1. Using estimates, which are familiar by now, we arrive at

$$|(\Phi, W(g_n, \kappa, \rho) \Phi)| \leq c_{21}(\rho + 1)^{-\delta/2} (\|N_\tau^{1/2} \Phi\|^2 + c_{22}n \|\Phi\|^2) \\ \leq \frac{\epsilon}{2} \|N_\tau^{1/2} \Phi\|^2 + c_{23}n \|\Phi\|^2 \quad (4.26)$$

for ρ sufficiently large.

We turn to a discussion of δM , which describes the mass-renormalization cancellation. The mass counterterm $-\frac{1}{2}\delta m^2(\kappa, \rho) \int : \varphi(x)^2 : g_n(x)^2 dx$ may be written as a sum of four terms

$$\int \beta(k_1, k_2) \{a(k_1)^* a(k_2)^* + a(k_1)^* a(-k_2) \\ + a(k_2)^* a(-k_1) + a(-k_1) a(-k_2)\} dk_1 dk_2 \quad (4.27)$$

with

$$\beta(k_1, k_2) = -\frac{1}{2}\delta m^2(\kappa, \rho)(4\pi)^{-1} (\mu_1\mu_2)^{-1/2} (\tilde{g}_n * \tilde{g}_n)(k_1 + k_2).$$

The corresponding terms in H_1 only differ by kernels $\gamma_i(k_1, k_2)$ ($i = 1 \cdots 4$) replacing $\beta(k_1, k_2)$. Here

$$\gamma_i(k_1, k_2) = - \sum_{\substack{|j| \leq n \\ |j'| \leq n}} e^{i(jk_1 + j'k_2)} \int w_{\kappa, \rho}^C(k_1, p_1, p_2) w_{\kappa, \rho}^C(k_2, -p_1, -p_2) \\ \times e^{i(j-j')(p_1 + p_2)} \delta_i dp_1 dp_2,$$

where

$$\begin{aligned}\delta_1 &= (\omega_1 + \omega_2 + \mu_2)^{-1} & \delta_2 &= (\omega_1 + \omega_2)^{-1} \\ \delta_3 &= (\omega_1 + \omega_2 + \mu_1)^{-1} & \delta_4 &= (\omega_1 + \omega_2 + \mu_2)^{-1}.\end{aligned}$$

Therefore we will be interested in the difference $(\beta - \gamma_i)(k_1, k_2)$. Now the results of Chapter 3 are not directly applicable, but again we may estimate with the help of localization indices. Define

$$\beta(k_1, k_2, J) = -\frac{\delta m^2(\kappa, \rho)}{2} (4\pi)^{-1} (\mu_1 \mu_2)^{-1/2} \int \tilde{h}(k_1 + \eta) \tilde{h}(k_2 - \eta) e^{i\eta J} d\eta,$$

$$\gamma_i(k_1, k_2, J) = -\int w_{\kappa, \rho}^C(k_1, p_2, p_2) w_{\kappa, \rho}^C(k_2, -p_1, -p_2) e^{iJ(p_1 + p_2)} \delta_i dp_1 dp_2,$$

such that

$$\begin{aligned}\beta(k_1, k_2) &= \sum_{\substack{|j| \leq n \\ |j'| \leq n}} \beta(k_1, k_2, j - j') e^{i(jk_1 + j'k_2)}, \\ \gamma_i(k_1, k_2) &= \sum_{\substack{|j| \leq n \\ |j'| \leq n}} \gamma_i(k_1, k_2, j - j') e^{i(jk_1 + j'k_2)}.\end{aligned}$$

We will first show that

$$\left\| \frac{\partial^{\alpha_1}}{\partial k_1^{\alpha_1}} \frac{\partial^{\alpha_2}}{\partial k_2^{\alpha_2}} (\beta - \gamma_i)(\cdot, J) \right\|_2 \leq c(\alpha_1, \alpha_2, m)(1 + \rho)^{-\delta} (|J| + 1)^{-m} \quad (4.28)$$

uniformly in κ for some $\delta > 0$.

It is sufficient to take $i = 1$, the other cases being similar. Set

$$\begin{aligned}\beta_1 &= 1; & \beta_2 &= \omega(\xi)^{-1}; & \beta_3 &= \chi_\kappa\left(0, \frac{\xi}{2}, -\frac{\xi}{2}\right); \\ \tilde{\beta}_1 &= \frac{1}{2} \frac{\omega_1 \omega_2 - p_1 p_2 - m_f^2}{\omega_1 \omega_2}; & \tilde{\beta}_2 &= \delta_1; & \tilde{\beta}_3 &= \chi_\kappa(k_1, p_1, p_2) \chi_\kappa(+k_2, -p_1, -p_2)\end{aligned}$$

such that

$$\begin{aligned}\beta(k_1, k_2, J) &= \frac{\lambda^2}{4\pi} (\mu_1 \mu_2)^{-1/2} \int \beta_1 \beta_2 \beta_3 \tilde{h}(k_1 + \eta) \tilde{h}(k_2 - \eta) e^{iJ\eta} (1 - \Theta_\rho(\xi))^2 d\xi d\eta, \\ \gamma_1(k_1, k_2, J) &= \frac{\lambda^2}{4\pi} (\mu_1 \mu_2)^{-1/2} \int \tilde{\beta}_1 \tilde{\beta}_2 \tilde{\beta}_3 \tilde{h}(k_1 + \eta) \tilde{h}(k_2 - \eta) e^{iJ\eta} (1 - \Theta_\rho(\xi))^2 d\xi d\eta.\end{aligned}$$

First we write

$$\beta_1 \beta_2 \beta_3 - \tilde{\beta}_1 \tilde{\beta}_2 \tilde{\beta}_3 = (\beta_1 - \tilde{\beta}_1) \tilde{\beta}_2 \tilde{\beta}_3 + \beta_1 (\beta_2 - \tilde{\beta}_2) \tilde{\beta}_3 + \beta_1 \beta_2 (\beta_3 - \tilde{\beta}_3).$$

Now

$$\begin{aligned}\beta_1 - \bar{\beta}_1 &= \frac{1}{2} \frac{\omega_1 \omega_2 + p_1 p_1 + m_f^2}{\omega_1 \omega_2} \\ &= \frac{1}{2} \left\{ \left[\omega \left(\frac{\xi + \eta}{2} \right) \omega \left(\frac{\xi - \eta}{2} \right) - \frac{1}{4} (\xi^2 - \eta^2) + m_f^2 \right] / \omega \left(\frac{\xi + \eta}{2} \right) \omega \left(\frac{\xi - \eta}{2} \right) \right\}\end{aligned}$$

so

$$\left| \frac{\partial^m}{\partial \eta^m} (\beta_1 - \bar{\beta}_1) \right| \leq c(m) \omega(\eta)^{1\delta} \omega(\xi)^{-2\delta} \quad (0 \leq \delta \leq \tfrac{1}{2}).$$

Also

$$\beta_2 - \bar{\beta}_2 = \omega(\xi)^{-1} (\omega_1 + \omega_2 + \mu_2)^{-1} (\mu_2 + \omega_1 + \omega_2 - \omega(\xi)),$$

so the estimate $|\omega_1 + \omega_2 - \omega(\xi)| \leq \text{const } \omega(\eta)$ easily gives

$$\left| \frac{\partial^{\alpha_2}}{\partial k_2^{\alpha_2}} \frac{\partial^m}{\partial \eta^m} (\beta_2 - \bar{\beta}_2) \right| \leq c(\alpha_2, m) \omega(\xi)^{-1-2\delta} (\omega(\eta) + \mu_2)^{2\delta}.$$

Finally

$$\begin{aligned}\left| \frac{\partial^\alpha}{\partial k^\alpha} \frac{\partial^m}{\partial \eta^m} (\omega_1 + \omega_2 + \mu)^{-1} \right| &\leq c(\alpha, m) \omega(\xi)^{-1}, \\ \left| \frac{\partial^\alpha}{\partial k^\alpha} \frac{\partial^m}{\partial \eta^m} \chi_\kappa(k, p_1, p_2) \right| &\leq c(\alpha, m) (\kappa + 1)^{-(\alpha+m-1)} \omega(\xi)^{-1} \\ &\quad (\text{for } \alpha + m \geq 1), \\ \left| \chi_\kappa(k, p_1, p_2) - \chi_\kappa \left(0, \frac{\xi}{2}, -\frac{\xi}{2} \right) \right| &\leq \text{const} (\kappa + 1)^{-\delta} (\omega(\eta) + \mu)^{3\delta} \omega(\xi)^{-2\delta}.\end{aligned}$$

(See the proof of Lemma 6.1.1 in [11]).

With these estimates it is easy to show that

$$\begin{aligned}\left| \frac{\partial^{\alpha_1}}{\partial k_1^{\alpha_1}} \frac{\partial^{\alpha_2}}{\partial k_2^{\alpha_2}} (\beta - \gamma_1)(k_1, k_2, J) \right| \\ \leq c(\alpha_1, \alpha_2, m) (1 + \rho)^{-\delta} (\mu_1 \mu_2)^{-1/2+3\delta} (|J| + 1)^{-m} (1 + |k_1 + k_2|)^{-1}\end{aligned}$$

uniformly in κ . This proves (4.28) if we choose $\delta < (1/12)$. To apply this, first set

$$\begin{aligned}\Gamma_i(k_1, k_2, J) &= (\beta - \gamma_i)(k_1, k_2, J); \\ \Gamma_{i,(j,j')}(k_1, k_2, J) &= e^{i(jk_1 + j'k_2)} \Gamma_i(k_1, k_2, J)\end{aligned}$$

such that corresponding to the sum in (4.27)

$$\begin{aligned}
 -\delta M &= - \sum_{i=1}^4 \delta M_i \\
 &= \sum_{i=1}^4 \sum_{|j|, |j'| \leq n} \left\{ \int \Gamma_{i,(j,j')}(k_1, k_2, j-j') \cdot a(k_1)^{\#} a(k_2)^{\#} \cdot dk_1 dk_2 \right\}, \quad (4.29)
 \end{aligned}$$

where $a(k)^{\#} = a(k)^{*}$ or $a(-k)$.

Now we localize $\Gamma_{i,(j,j')}(k_1, k_2, j-j')$ in the variables k_1, k_2 , i.e., we define [see (3.8)]

$$\begin{aligned}
 \Gamma_{i,(j,j');(l_1,l_2)}(k_1, k_2, j-j') &= e^{i(jk_1+j'k_2)} \int \Gamma_i(k_1 - k_1', k_2 - k_2', j-j') \\
 &\quad \times e^{i(l_1-j)k_1' + i(l_2-j')k_2'} \tilde{\eta}(k_1') \tilde{\eta}(k_2') dk_1' dk_2'.
 \end{aligned}$$

The proof of Lemma 3.1 combined with the estimate (4.28) gives uniformly in κ

$$\begin{aligned}
 \| \Gamma_{i,(j,j');(l_1,l_2)}(\cdot, j-j') \|_2 &\leq c'(\alpha_1, \alpha_2, m)(1+\rho)^{-\delta} (|l_1-j|+1)^{-\alpha_1} \\
 &\quad \times (|l_2-j'|+1)^{-\alpha_2} (|j-j'|+1)^{-m}. \quad (4.30)
 \end{aligned}$$

We will only need (4.30) with $\alpha_1 = \alpha_2 = m = 4$.

For $\Phi \in D$ we have

$$\begin{aligned}
 \left| \sum_{i=1}^4 (\Phi, \delta M_i \Phi) \right| &\leq \sum_{i=1}^4 \sum_{\substack{|j|, |j'| \leq n \\ l_1, l_2 \in \mathbb{Z}}} \left| \left(\Phi, \int \Gamma_{i,(j,j');(l_1,l_2)}(k_1, k_2, j-j') \right. \right. \\
 &\quad \left. \left. \times \cdot a(k_1)^{\#} a(k_2)^{\#} \cdot dk_1 dk_2 \Phi \right) \right|
 \end{aligned}$$

so by Lemma 2.9

$$\begin{aligned}
 \left| \sum_{i=1}^n (\Phi, \delta M_i \Phi) \right| &\leq c_{24}(1+\rho)^{-\delta} \sum_{\substack{l_1, l_2 \in \mathbb{Z} \\ |j| \leq n}} \| (N_{0,l_1}^0 + 1)^{1/2} \Phi \| \| (N_{0,l_2}^0 + 1)^{1/2} \Phi \| \\
 &\quad \times (|l_1-j|+1)^{-4} (|l_2-j|+1)^{-2} \\
 &\leq c_{25}(1+\rho)^{-\delta} (\| N^{1/2} \Phi \|^2 + c_{26} n \| \Phi \|^2).
 \end{aligned}$$

This completes the discussion of $(H_1 - H)(g_n, \kappa, \rho)$ for large ρ .

We turn to a discussion of H_2 . In the notation of [11] H_2 may be written as a sum of Wick-ordered terms

$$H_2 = \sum_{i=1}^{16} Z_i,$$

where each Z_i is of the form (3.1a) for $1 \leq i \leq 4$ and of the form (3.1b for $i > 4$. For most of the terms, we only outline the estimates, since the essential arguments already have been explained.

First the κ -independence is guaranteed by Lemma 4.1 and the ρ -dependence is controlled by Lemma 4.2. In particular $Z_1 \cdots Z_4$ may be discussed in a way similar to $W(g_n, \kappa, \rho)$, since the discussion in [11] shows that Lemma 3.1 may be applied. $Z_5 \cdots Z_8$ are discussed as the terms δM_i in δM : It is easy to see that their kernels may be written in such a way that Lemma 3.3 is applicable, thus substituting Estimate (4.30). Z_9 is a real number bounded by $\text{const } n$ [see Estimate (4.21)]. For Z_{10} and Z_{11} again Lemma 3.3 may be used. We turn to Z_{12} which is a bit more delicate since it contains four fermions and Lemma 3.3 is not directly applicable. However, we may argue as follows:

For $\Phi \in D$, by definition of Z_{12} in [11]

$$(\Phi, Z_{12}\Phi) = \int \|Z_{17}(k) \Phi\|^2 dk,$$

$$Z_{17}(k) = \bar{\mu}(k)^{1/2} \sum_{|j| \leq n} \int b(-p_1) b'(-p_2) w_{\kappa, \rho, j}^C(k, p_1, p_2) (\omega_1 + \omega_2)^{-1} dp_1 dp_2,$$

$$\bar{\mu}(k) = \bar{\mu}(k, 0) = \mu(k) - c\mu(k)^\tau.$$

Now define

$$v_{\kappa, \rho}(k, p_1, p_2) = \bar{\mu}(k)^{1/2} w_{\kappa, \rho}^C(k, p_1, p_2);$$

$$v_\rho(k, p_1, p_2) = \bar{\mu}(k)^{1/2} w^C(k, p_1, p_2)(1 - \Theta_\rho(\xi)),$$

$$v_{\kappa, \rho, j}(k, p_1, p_2) = e^{ij(p_1 + p_2)} v_{\kappa, \rho}(k, p_1, p_2),$$

i.e., $v_{\kappa, \rho, j}$ is the translate of $v_{\kappa, \rho}$ only in the fermion variables. Set

$$Z_{17, j}(k) = \int v_{\kappa, \rho, j}(k, p_1, p_2) b(-p_1) b'(-p_2) dp_1 dp_2;$$

thus

$$(\Phi, Z_{12}\Phi) = \sum_{|j|, |j'| \leq n} \left(\Phi, \int Z_{17, j'}(k)^* Z_{17, j}(k) e^{i(j-j')k} dk \Phi \right).$$

Next we localize $v_{\kappa,\rho,j}(k, p_1, p_2)$ in the variables p_1, p_2 , obtaining the kernel $v_{\kappa,\rho,j,(l_1,l_2)}(k, p_1, p_2)$ and the corresponding operator $Z_{17,j,(l_1,l_2)}(k)$ such that

$$(\Phi, Z_{12}\Phi) = \sum_{\substack{|j| \leq n \\ |j'| \leq n}} \sum_{\substack{l_1', l_2' \\ l_1', l_2'}} \left(\Phi, \int Z_{17,j',(l_1',l_2')}(k)^* Z_{17,j,(l_1,l_2)}(k) e^{i(j-j')k} dk \Phi \right)$$

and hence by Schwarz inequality

$$\begin{aligned} 0 &\leq (\Phi, Z_{12}\Phi) \\ &\leq c_{28} \sum_{\substack{|j| \leq n \\ |j'| \leq n}} \sum_{\substack{l_1', l_2' \\ l_1', l_2'}} (1 + |j - j'|)^{-2} \sup_{\alpha + \alpha' \leq 2} \left(\int \left\| \frac{\partial^{\alpha'}}{\partial k^{\alpha'}} Z_{17,j',(l_1',l_2')}(k) \Phi \right\| dk \right)^{1/2} \\ &\quad \times \left(\int \left\| \frac{\partial^{\alpha}}{\partial k^{\alpha}} Z_{17,j,(l_1,l_2)}(k) \Phi \right\|^2 dk \right)^{1/2}. \end{aligned}$$

Now $\omega_1^{-\tau/2} \omega(\xi)^{\delta} D^{\alpha(3)} v_{\rho} \in L^2(\mathbb{R}^3)$ for all $\alpha(3)$. Taking the L^2 -norm with respect to the variables p_1, p_2 , the arguments used in Lemmas 3.1, 4.1, 4.2 show that

$$\begin{aligned} &\left\| \tilde{\omega}_1^{-\tau/2} \frac{\partial^{\alpha}}{\partial k^{\alpha}} v_{\kappa,\rho,j,(l_1,l_2)} \right\|_2 \\ &\leq C(\alpha_1, \alpha_2, \alpha)(1 + \rho)^{-\delta} \mu^{-(1+\tau-2\delta)/2} (|l_1 - j| + 1)^{-\alpha_1} (|l_2 - j| + 1)^{-\alpha_2} \end{aligned}$$

uniformly in κ for some small $\delta > 0$. Hence the localized N_{τ} estimate in form of Lemma 2.4 with $|B| = 1$ gives with $2\alpha = \alpha_1 = \alpha_2 = 4$

$$\begin{aligned} 0 &\leq (\Phi, Z_{12}\Phi) \\ &\leq c_{29}(1 + \rho)^{-\delta} \sum_{\substack{|j| \leq n \\ |j'| \leq n}} \sum_{\substack{l_1', l_2' \\ l_1', l_2'}} (1 + |j - j'|)^{-2} (|l_1 - j| + 1)^{-4} (|l_2 - j| + 1)^{-2} \\ &\quad \times (|l_1' - j'| + 1)^{-4} (|l_2' - j'| + 1)^{-2} \|(N_{\tau,l_1}^+ + 1)^{1/2} \Phi\| \|(N_{\tau,l_1'}^+ + 1)^{1/2} \Phi\| \end{aligned}$$

and this again gives

$$\begin{aligned} 0 &\leq (\Phi, Z_{12}\Phi) \\ &\leq c_{30}(1 + \rho)^{-\delta} (\|N_{\tau}^{1/2}\Phi\|^2 + c_{31}n \|\Phi\|^2) \\ &\leq \epsilon \|N_{\tau}^{1/2}\Phi\|^2 + c_{32}n \|\Phi\|^2 \quad \text{uniformly in } \kappa \text{ for large } \rho. \end{aligned}$$

The terms $Z_{13}, \dots, Z_{16}$ contain two boson and two fermion operators each. As in [11] we write their sum as a negative operator plus two Wick-ordered monomials of the form (3.1b) containing only two fermion operators each. Their kernels again satisfy the conditions of Lemma 3.3 such that localized N_{τ} estimates may be

applied to give $Z_{13} + \cdots Z_{16} \leq \epsilon N_r + nc_{33}$ uniformly in κ on $D \times D$ for large ρ . Combining these estimates finally proves Lemma 4.3 and thus the first-order volume-dependent N_r estimate.

5. THE UPPER BOUND FOR THE VACUUM ENERGY

In perturbation theory the vacuum energy is formally obtained from an expression which is the sum of all possible connected diagrams without external legs of the theory considered (see e.g. [18, Chapter IV]), each of which behaves like $m(g)$ in g . Although this perturbation series is supposed to diverge in our case, nevertheless it gives us an idea of how to prove the upper bound (1.5). We look for an approximate unitary-dressing transformation $U(g) = \exp -A(g)$, where $A(g)$ is a simple expression in terms of annihilation and creation operators. Then $U(g)$ applied to the free vacuum is a reasonably good approximation to the ground state of $H(g)$ and we expect the vacuum expectation value $\langle U(g)^{-1} H(g) U(g) \rangle_0$ to behave like $m(g)$. Since $E(g) = \inf \text{spectrum } H(g) \leq \langle U(g)^{-1} H(g) U(g) \rangle_0$, this would prove the upper bound (1.5). Of course it is sufficient to prove corresponding estimates for $H(g, \kappa)$, which are uniform in κ , since $H(g)$ is the uniform resolvent limit of $H(g, \kappa)$. Thus we have to find a suitable $U(g, \kappa) = \exp -A(g, \kappa)$ and we take $A(g, \kappa)$ to be of the form as suggested by P. Federbush and B. Gidas [3]. The first thing we have to do is to expand

$$\exp A(g, \kappa) H(g, \kappa) \exp -A(g, \kappa) = \sum_{n=0}^{\infty} \frac{(\text{ad } A(g, \kappa))^n}{n!} H(g, \kappa)$$

as a sum of connected Friedrichs' diagrams. In lowest order the infinities related to the renormalization cancel and in higher orders they do not occur at all. Then we estimate the number of vacuum diagrams occurring in each order and show that this number is bounded by $(\text{const})^n (n!)^{1/2}$. This strongly suggests that a summation is possible, if we obtain uniform estimates for the individual vacuum diagrams. Now it is easy to verify that each individual vacuum graph G occurring in the above expansion is bounded in absolute value by $c(n) m(g)$. Indeed the g -dependence of G is given by

$$G = \int \prod_{i=1}^n g(x_i) G_{\kappa}(x_1 - x_n, \cdots x_{n-1} - x_n) dx_1 \cdots dx_n,$$

where n is the order and where G_{κ} does not depend on g . We will prove that G_{κ} is of the form $\prod_{i=1}^{n-1} (m_i^2 - (\partial^2/\partial x_i^2)^3 \tilde{G}_{\kappa})$, with $\tilde{G}_{\kappa} \in L^1(\mathbb{R}^{n-1})$ and a bound which is uniform in κ . Now in order to do the summation over all graphs, we will actually show that this bound is such that the above $c(n)$ behaves like $(\text{const})^n$ with a

constant independent of n and κ . The proof of this estimate uses the kernel theorem of J. Glimm [7], by which a large graph is estimated by splitting it into smaller graphs of bounded size and then using Schwarz inequality. However, it is easy to see that this method is not directly applicable to G , since this would give unwanted powers in $m(g)$. In order to avoid this, we introduce a graph G_κ^μ which is obtained in a simple way from G and whose form is suggested by considering $\tilde{g}$ as an approximate δ -function. In particular G_κ^μ is not g -dependent and we may estimate G_κ^μ with help of Glimm's kernel theorem and the estimates for $\tilde{G}_\kappa$ of the desired form follow.

We now turn to a proof of these statements. In this section we assume again $g \in \mathfrak{G}(\alpha)$. χ_κ is as in Chapter 4. In particular $\chi_\kappa \in \mathfrak{S}(\mathbb{R}^3)$ for fixed κ . With a slight change in notation we set

$$w_\kappa^c(k, p_1, p_2) = (\lambda/4\pi) \tilde{g}(k + p_1 + p_2)(\mu\omega_1\omega_2)^{-1/2} (\omega_1\omega_2 - p_1 p_2 - m_f^2)^{1/2} \text{sign}(p_1 - p_2) \chi_\kappa(k, p_1, p_2). \quad (5.1)$$

Define

$$A(g, \kappa) = \int w_\kappa^c(k, p_1, p_2)(\mu + \omega_1 + \omega_2)^{-1} \varphi(k)(b(p_1)^* b'(p_2)^* - b(-p_1) b'(-p_2)) dk dp_1 dp_2, \quad (5.2)$$

where

$$\varphi(k) = a(k)^* + a(-k), \quad (5.3)$$

such that (see Proposition 1.2.3c in [11]) for fixed κ and g

$$\|A(g, \kappa) \Phi\| \leq \text{const} \|N^{1/2} \Phi\| \text{ for } \Phi \text{ in Fock space.} \quad (5.4)$$

Hence in particular D is a dense set of analytic vectors for $A(g, \kappa)$. Also $A(g, \kappa)^* = -A(g, \kappa)$ on D , so $A(g, \kappa)$ is essentially skew-adjoint on D by Nelson's theorem and $U(g, \kappa) = \exp -A(g, \kappa)$ is unitary. By the same arguments

$$\exp A(g, \kappa) H(g, \kappa) \exp -A(g, \kappa) = \sum_{n=0}^{\infty} (n!)^{-1} (\text{ad } A(g, \kappa))^n H(g, \kappa) \quad (5.5)$$

as an equality between bilinear forms on $D \times D$. Let us study the right-hand side in more detail. Expanding the terms into Wick-ordered monomials, we first note that the corresponding Friedrichs diagrams [4] are connected. Secondly [see (1.7), (4.7)]

$$[\mathfrak{c}(g, \kappa), A(g, \kappa)] = [:(P(\varphi)(g):, A(g, \kappa)] = 0$$

since

$$[\varphi(k), \varphi(k')] = 0$$

for all k, k' . Thus

$$(\text{ad } A(g, \kappa))^n H(g, \kappa) = -(\text{ad } A(g, \kappa))^{n-1} \dot{A}(g, \kappa) + (\text{ad } A(g, \kappa))^n H_I(g, \kappa) \quad (5.6)$$

with

$$\dot{A}(g, \kappa) = \text{ad } H_0 A(g, \kappa) = [H_0, A(g, \kappa)].$$

Since [see (4.9)]

$$\langle \text{ad } A(g, \kappa) H_I(g, \kappa) \rangle_0 = \frac{1}{2} \langle \text{ad } A(g, \kappa) \dot{A}(g, \kappa) \rangle_0 = E(g, \kappa), \quad (5.7)$$

the energy-renormalization counter term is cancelled. Thus

$$\begin{aligned} & \langle \exp A(g, \kappa) H(g, \kappa) \exp -A(g, \kappa) \rangle_0 \\ &= \sum_{n=2}^{\infty} \left\{ \frac{1}{n!} \langle \text{ad } A(g, \kappa))^n H_I(g, \kappa) \rangle_0 - \frac{1}{(n+1)!} \langle \text{ad } A(g, \kappa))^n \dot{A}(g, \kappa) \rangle_0 \right\}. \end{aligned} \quad (5.8a)$$

It is easy to see that terms with n even vanish. Also $W(g, \kappa)$ [see (4.6)] gives no contribution and

$$\langle \exp A(g, \kappa); P(\varphi)(g); \exp -A(g, \kappa) \rangle_0 = 0. \quad (5.8b)$$

Next we need a definition. Let A_1 and A_2 be Wick-ordered monomials. Then a Wick term is a Wick-ordered monomial appearing in the Wick ordering of $A_1 A_2$ whose numerical kernel is (up to a sign) the product of the numerical kernels of A_1 and A_2 with variables belonging to paired operators identified and integrated over. This definition easily extends to the case when we have to deal with a sum of products of an arbitrary number of Wick-ordered monomials. (Thus a Wick term never contains combinatorial factors arising from possible different pairing schemes.)

The following lemma will be useful and is easy to prove:

LEMMA 5.1. *Let A_1 (resp. A_2) be a Wick-ordered monomial with n_1 (resp. n_2) creation and annihilation operators. Then $A_1 A_2$ (resp. $[A_1, A_2]$) may be written as a sum of at most $(n_2 + 1)^{n_1}$ Wick terms, each having at most $n_1 + n_2$ (resp. $n_1 + n_2 - 2$) creation and annihilation operators.*

Now let us look at the Wick terms appearing in the Wick ordering of the expressions $(\text{ad } A(g, \kappa))^n H_I(g, \kappa)$ or $(\text{ad } A(g, \kappa))^n \dot{A}(g, \kappa)$. By definition the order of these expressions is $n + 1$. We will be interested in the number of Wick terms occurring. Consider $(\text{ad } A(g, \kappa))^n H_I(g, \kappa)$ first. Since $H_I(g, \kappa)$ as well as $A(g, \kappa)$

contain the boson operators only in the combination $\varphi(k) = a(k)^* + a(-k)$, the ad-operation only gives contractions from the fermion parts. In particular $\text{ad } A(g, \kappa)^n H_I(g, \kappa)$ is of the form

$$(\text{ad } A(g, \kappa))^n H_I(g, \kappa) = \int \varphi(k_1) \cdots \varphi(k_{n+1}) W_n(k_1 \cdots k_{n+1}) dk_1 \cdots dk_{n+1}, \quad (5.9)$$

where $W_n(k_1 \cdots k_n)$ for fixed $k_1 \cdots k_{n+1}$ is a sum of Wick-ordered monomials in the fermion operators. The number of fermions is at most two. To see this, we proceed by induction. The case $n = 0$ is trivial by definition of $H_I(g, \kappa)$. Assuming the statement to be true for n , we have

$$(\text{ad } A(g, \kappa))^{n+1} H_I(g, \kappa) = \int \varphi(k_1) \cdots \varphi(k_{n+1}) [A(g, \kappa), W_n(k_1 \cdots k_{n+1})] dk_1 \cdots dk_{n+1}$$

and the assertion for $n + 1$ follows from Lemma 5.1, since each of the two Wick-ordered monomials in $A(g, \kappa)$ has two fermions. This inductive proof also shows that the number of different Wick terms contributing to W_n is at most 4.18^n , since again by Lemma 5.1 nine is the maximal number of Wick terms arising from the commutator of two Wick-ordered monomials containing at most two fermions each. Also the factor 4 comes from the 4 summands in $H_I(g, \kappa)$, if $H_I(g, \kappa)$ is written in terms of the operator $\varphi(k)$ and Wick-ordered monomials in the fermions.

We turn to the Wick ordering of $\varphi(k_1) \cdots \varphi(k_{n+1})$. By induction it is easy to see that Wick ordering gives at most $(n + 1)!$ Wick terms. However, we will be mostly interested in another bound: The number of Wick terms having l_1 creation and l_2 annihilation operators is bounded by

$$[(n + 1)!(n + 1 - l_1 - l_2)!/l_1!l_2!(n + 1 - l_1 - l_2)!]$$

if $n + 1 - l_1 - l_2$ is even and zero else, since $(2m)!! = \prod_{i=0}^{m-1} (2i + 1)$ is the number of different pairings of $2m$ elements. For $l_1 = l_2 = 0$ the above number is $(n + 1)!!$ if n is odd and zero else, so the number of vacuum diagrams ($l_1 = l_2 = 0$) occuring in $(\text{ad } A(g, \kappa))^n H_I(g, \kappa)$ is bounded by $4.18^n(n + 1)!!$ for n odd and zero else. Note that

$$\begin{aligned} (n + 1)!! &= \prod_{i=0}^{(n-1)/2} (n + 1 - 2i - 1) \leq \prod_{i=0}^{(n-1)/2} (n + 1 - 2i) \\ &= 2^{(n+1)/2} \prod_{i=0}^{(n-1)/2} \left(\frac{n+1}{2} - i \right) = 2^{(n+1)/2} \left(\frac{n+1}{2} \right)!. \end{aligned}$$

We turn to $(\text{ad } A(g, \kappa))^n \dot{A}(g, \kappa)$. There the situation is slightly more complicated

since now boson contractions are possible. However, we have the following representation

$$\begin{aligned}
 (\text{ad } A(g, \kappa))^n \dot{A}(g, \kappa) &= \int \varphi(k_1) \cdots \varphi(k_n) \dot{\varphi}(k_{n+1}) W_n^1(k_1 \cdots k_{n+1}) dk_1 \cdots dk_{n+1} \\
 &\quad + \int \varphi(k_1) \cdots \varphi(k_{n+1}) W_n^2(k_1 \cdots k_{n+1}) dk_1 \cdots dk_{n+1} \\
 &\quad + \int \varphi(k_1) \cdots \varphi(k_{n-1}) W_n^3(k_1 \cdots k_{n-1}) dk_1 \cdots dk_{n-1} \\
 &= W_n^1 + W_n^2 + W_n^3,
 \end{aligned}$$

where W_n^1 (resp. W_n^2 , resp. W_n^3) is a sum of Wick-ordered monomials in the fermion operators, each containing at most two (resp. two, resp. four) fermions. Also

$$\dot{\varphi}(k) = \mu(k)(a(k)^* - a(-k)), [\varphi(k), \dot{\varphi}(k')] = 2\delta(k + k')\mu(k).$$

The proof is again by induction. For $n = 0$ it is easy to see that $A(g, \kappa)$ may be written in the form (5.10) since $[H_0, \varphi(k)] = \dot{\varphi}(k)$. Assume the theorem to be true for n . By Lemma 5.1 $\text{ad } A(g, \kappa) W_n^i$ is an operator of the form W_{n+1}^i ($i = 2, 3$) and $\text{ad } A(g, \kappa) W_n^1$ is the sum of an operator of the form W_{n+1}^1 (if only fermions are contracted) or W_{n+1}^3 (if bosons are contracted). Let $|w_n^i|$ be the number of different Wick terms contributing to W_n^i ($i = 1, 2, 3$). Since $|w_0^1| = |w_0^2| = 2$, the above arguments and Lemma 5.1 give the estimates

$$\begin{aligned}
 |w_n^1| &\leq 2 \cdot 18^n, & |w_n^2| &= 2 \cdot 18^n, \\
 |w_n^3| &\leq 2 \cdot 9 |w_{n-1}^1| + 2 \cdot 25 |w_{n-1}^3| \leq 50(|w_{n-1}^1| + |w_{n-1}^3|) \\
 &\leq 2n \cdot 50^n 10^n = 2n(500)^n.
 \end{aligned} \tag{5.11}$$

In particular the number of vacuum diagrams occurring in the Wick-ordered expansion of $\text{ad } A(g, \kappa)^n \dot{A}(g, \kappa)$ is bounded by $4(n+1)(500)^n (n+1)!!$ if n is odd and zero if n is even. This completes the estimate on the number of contributions of vacuum diagrams to (5.8). We now turn to the discussion of the individual vacuum diagrams. What we have in mind is to use Glimm's kernel theorem [7] which gives uniform estimates for diagrams, and we will refer to the clear elucidation of this theorem given by K. Osterwalder [14], who made systematic use of Weinberg's power-counting theorem [20]. We will show that each vacuum diagram occurring in (5.8a) is bounded in absolute value by $K^{n+1}m(g)$ where $0 < K < \infty$ does not depend on κ and the order $n+1$.

Now let G be a vacuum diagram of $(n+1)^{\text{st}}$ order ($n \geq 3$, n odd) occurring in

(5.8a). G will also denote its numerical value. We say G is of type I or II according to the following rule. Set

$$\begin{aligned} \dot{A}(g, \kappa) &= \int \frac{\omega_1 + \omega_2}{(\mu + \omega_1 + \omega_2)} w_\kappa^C(k, p_1, p_2) \varphi(k) \\ &\quad \times \{b(p_1)^* b'(p_2)^* + b(-p_1) b'(-p_2)\} dk dp_1 dp_2 \\ &\quad + \int \frac{1}{(\mu + \omega_1 + \omega_2)} w_\kappa^C(k, p_1, p_2) \dot{\varphi}(k) \\ &\quad \times \{b(p_1)^* b'(p_2)^* - b(-p_1) b'(-p_2)\} dk dp_1 dp_2 \\ &= \dot{A}_1(g, \kappa) + \dot{A}_2(g, \kappa). \end{aligned}$$

Then G is of type I, if G appears in the expansion of $(\text{ad } A(g, \kappa))^n H_{I, \kappa}(g)$ or $(\text{ad } A(g, \kappa))^n \dot{A}_1(g, \kappa)$ and of type II if G appears in the expansion of

$$(\text{ad } A(g, \kappa))^n \dot{A}_2(g, \kappa).$$

In particular, if G is of type I, then G remains connected after deleting all $(n+1)/2$ boson lines, and we say G is fermion connected. In any case, G contains $(3/2)(n+1)$ internal lines, giving rise to the same number of integrations. Let $k_i, p_{1,i}, p_{2,i}$ ($i = 1, \dots, n+1$) be the momenta of the particles at the i -th vertex. Set $l_i = k_i + p_{1,i} + p_{2,i}$. Since G is connected, $\sum_{i=1}^{n+1} l_i = 0$ is the only nontrivial relation between the l_i . Now choose $q_1 \cdots q_{(n+3)/2}$ as independent loop momenta such that all $p_{1,i}, p_{2,i}$ ($i = 1, \dots, n+1$) are linear combinations of the q_j ($j = 1, \dots, n+1$). This is possible due to the following consideration: If we set $l_i = 0$ ($i = 1, \dots, n+1$), then we have momentum conservation at the vertices and all boson momenta k_i are replaced by $-p_{1,i} - p_{2,i}$. In particular we have the following representation for G ;

$$G = \int \prod_{i=1}^{n+1} \tilde{g}(l_i) \delta\left(\sum_{i=1}^{n+1} l_i\right) G_\kappa(l_1 \cdots l_{n+1}; q_1 \cdots q_{(n+3)/2}) dl_1 \cdots dl_{n+1} dq_1 \cdots dq_{(n+3)/2}. \quad (5.12)$$

Now

$$G'_\kappa = \int G_\kappa(0 \cdots 0; q_1 \cdots q_{(n+3)/2}) dq_1 \cdots dq_{(n+3)/2} \quad (5.13)$$

is the value of the same diagram but with momentum conservation at all vertices. We will first prove that for some $K_1 > 0$

$$|G'_\kappa| < (K_1)^{n+1} \quad (5.14)$$

for all n and all vacuum diagrams G of order $(n+1)$ uniformly in κ .

To show this define

$$G_{\kappa}^{\mu} = \int \prod_{i=1}^{n+1} \mu(l_i)^{-2} \delta \left(\sum_{i=1}^{n+1} l_i \right) | G_{\kappa}(0 \cdots 0; q_1 \cdots q_{(n+3)/2}) | dq_1 \cdots dq_{(n+3)/2} dl_1 \cdots dl_{n+1}. \quad (5.15)$$

Obviously G_{κ}^{μ} corresponds to a graph having the same structure as G , however with a new numerical kernel. In particular, the cut-off function $\tilde{g}(l_i)$ at the i -th vertex has been replaced by $\mu(l_i)^{-2}$. More precisely G_{κ}^{μ} may be obtained from G in the following way: First let us reintroduce the momenta

$$k_i, p_{1,i}, p_{2,i} \quad (i = 1, \dots, n+1).$$

Then if $\tilde{g}(k_i + p_{1,i} + p_{2,i}) V_i(k_i, p_{1,i}, p_{2,i})$ was the vertex function of G at the i -th vertex, $\mu(k_i + p_{1,i} + p_{2,i})^{-2} | V_i(-p_{1,i} - p_{2,i}, p_{1,i}, p_{2,i}) |$ is the vertex function of G_{κ}^{μ} at the i -th vertex, such that the numerical kernel g_{κ}^{μ} of G_{κ}^{μ} is simply

$$g_{\kappa}^{\mu} = \prod_{i=1}^{n+1} \mu(k_i + p_{1,i} + p_{2,i})^{-2} | V_i(-p_{1,i} - p_{2,i}, p_{1,i}, p_{2,i}) |.$$

Now since

$$\begin{aligned} \int \prod_{i=1}^{n+1} \mu(l_i)^{-2} \delta \left(\sum_{i=1}^{n+1} l_i \right) dl_1 \cdots dl_{n+1} &= \int_{-\infty}^{+\infty} \tilde{\mu}^{-2}(x)^{n+1} dx \\ &= \int_{-\infty}^{+\infty} \left(\frac{\pi}{2m_b^2} \right)^{(n+1)/2} e^{-m_b(n+1)|x|} dx \\ &= \left(\frac{\pi}{2m_b^2} \right)^{(n+1)/2} \frac{2}{m_b(n+1)} \geqslant (K_2)^{n+1} \end{aligned}$$

for some $K_2 > 0$, we obviously have

$$| G_{\kappa}' | \leqslant | G_{\kappa}^{\mu} | (K_2)^{-n-1}$$

and hence it is sufficient to prove

$$| G_{\kappa}^{\mu} | \leqslant (K_3)^{n+1} \quad (5.16)$$

for some $K_3 > 0$ independent of κ .

We decompose G_{κ}^{μ} into subgraphs $G_0, G_1, \dots, G_k$ ($k = n-2$ if G is of type I, $k = n-1$ if G is of type II) in the following way. First we construct G_0 . If G is of type I, let Z denote the vertex in G coming from the factor $H_I(g, \kappa)$ (resp. $\hat{A}_1(g, \kappa)$) and let P_1 and P_2 be the two different vertices which are connected to Z by a fermion line. The existence of P_1 and P_2 is guaranteed by the fermion connectivity of G .

If G is of type II, let P_3 be the vertex corresponding to $\hat{A}_2(g, \kappa)$ and let P_4 be the vertex to which P_4 is connected by a boson line. Call this boson line the Z -line. Now we define G_0 with kernel g_0 to be the subgraph of G_κ^μ having Z, P_1, P_2 (resp. P_3, P_4) as vertices, having the lines of G_κ^μ which connect Z, P_1, P_2 (resp. P_3, P_4) as internal lines, and having all other lines in G_κ^μ originating from A, P_1 or P_2 (resp. P_3 or P_4) as external lines. Obviously the number of possible G_0 is finite.

Finally let $G_i (i = 1 \cdots k)$ with numerical kernel g_i denote the subgraphs of G_κ^μ , each of which consists of exactly one of the remaining vertices and each having as external lines all lines leaving the vertex. Since this vertex of G_i comes from a factor of $A(g, \kappa)$ [whose numerical kernel has an extra factor $(\mu + \omega_1 + \omega_2)^{-1}$,] the definition of G_κ^μ shows that each g_i is square-integrable uniformly for all $\kappa \leq \infty$. Hence by Schwarz inequality {see e.g. [7] Formula (2.2.3)}

$$\|G_\kappa^\mu\| \leq \prod_{i=1}^k \|g_i\|_2 \left\| \int_{\text{int}} g_0 \right\|_2 \leq (K_4)^{n-1} \left\| \int_{\text{int}} g_0 \right\|_2 \quad (k = n - 2 \text{ or } n - 1) \quad (5.17)$$

uniformly in κ for some $K_4 > 0$.

(We note again that it was in order to use an inequality of this form that we discuss G_κ^μ instead of G_κ' .) Equation (5.17) shows that we only have to evaluate $\left\| \int_{\text{int}} g_0 \right\|_2$ and show that it is finite for all G_0 . Here $\left\| \int_{\text{int}} g_0 \right\|_2$ means: Integrate g_0 over the internal variables of G_0 , and then take the L^2 -norm with respect to the remaining external variables of G_0 .

As in [14] we note that $\left\| \int_{\text{int}} g_0 \right\|_2^2$ is the value of a graph $(G_0 G_0^*)$ having no external lines, which is obtained by taking G_0 and the "mirror graph" G_0^* (obtained by taking the adjoint of the operator expression for G_0) and connecting pairwise the corresponding external lines. In particular, if G is of type I (resp. II), $(G_0 G_0^*)$ has two Z -vertices (resp. two Z -lines) and altogether 6 (resp. 4) vertices. Finally $(G_0 G_0^*)$ has the following essential structure, if G is of type I and hence fermion connected.

(S₁) Each vertex which is not a Z -vertex cannot have both its fermion lines connected to Z -vertices.

(S₂) The two Z -vertices are not connected by a fermion line.

The following lemma is easily obtained from the discussion in [14] [see in particular Eqs. (6.39) and (6.41) there]

LEMMA 5.2. $\left\| \int_{\text{int}} g_0 \right\|_2$ is finite for all κ ($\kappa \leq \infty$) if

$$d_H = -i_b(H) - v(H) + Z(H) + l(H) < 0 \quad (5.18)$$

for every subgraph H of $(G_0 G_0^*)$ having more than one vertex.

The notation we use is the following: d_H is called the superficial divergence of H and

$$\begin{aligned}
 i_b(H) &= \text{number of internal boson lines in } H; \\
 i(H) &= \text{number of internal lines in } H; \\
 v(H) &= \text{number of vertices in } H; \\
 Z(H) &= \begin{cases} \text{number of } Z\text{-vertices in } H, \text{ if } G \text{ is of type I,} \\ \text{number of } Z\text{-lines in } H, \text{ if } G \text{ is of type II;} \end{cases} \\
 l(H) &= \text{number of independent loops in } H.
 \end{aligned} \tag{5.19}$$

In particular $0 \leq Z(H) \leq 2$ and $Z(H) = i_b(H)$, if G is of type II.

The origin of $Z(H)$ in (5.18) is as follows: If G is of type I, every Z -vertex has a factor $(\mu + \omega_1 + \omega_2)^{-1}$ missing (or an additional factor $(\omega_1 + \omega_2)$ [see Definition of $\hat{A}_1(g, \kappa)$]). If G is of type II, every Z -line has an additional factor μ , namely the one appearing in $\hat{\phi}(k)$ [see definition of $\hat{A}_2(g, \kappa)$].

Since

$$l(H) = i(H) - v(H) + 1 \tag{5.20}$$

we obtain the condition

$$d_H = -i_b(H) + i(H) - 2v(H) + Z(H) + 1 < 0. \tag{5.21}$$

Also each vertex in H is endpoint of at most 2 fermion lines contained in $i(H)$ and thus

$$i(H) - i_b(H) \leq v(H). \tag{5.22}$$

Let us check (5.21), when G is of type I

- (α) $v(H) \geq 4$, then $d_H \leq -v(H) + 3 \leq -1 < 0$;
- (β) $v(H) = 3$;
- (β_1) $Z(H) \leq 1$, then $d_H \leq -v(H) + 2 = -1 < 0$;
- (β_2) $Z(H) = 2$, then $i(H) - i_b(H) < v(H) - 1$ by (S_1) and (S_2) and hence again $d_H < -v(H) + 3 - 1 = -1 < 0$,
- (γ) $v(H) = 2$;
- (γ_1) $Z(H) = 0$, then $d_H \leq -v(H) + 1 = -1 < 0$;
- (γ_2) $Z(H) = 1$, then $i(H) - i_b(H) \leq 1$ by (S_1) and

$$d_H \leq 1 - 2v(H) + 2 = -1 < 0;$$

- (γ_3) $Z(H) = 2$, then $i(H) = i_b(H)$ by (S_2) and

$$d_H = -2v(H) + 3 = -1 < 0.$$

If G is of type II, the cases (β_2) and (γ_3) are excluded since by construction of G_0 , $Z(H) = 2$ implies $v(H) = 4$. Also in the case (γ_2) again $i(H) - i_b(H) \leq 1$,

since otherwise G would not be connected. This concludes the proof of Estimate (5.17) and hence also of Estimate (5.14).

Now it is easy to estimate

$$G_{\kappa}'(l_1 \cdots l_{n+1}) = \int G_{\kappa}(l_1 \cdots l_{n+1}, q_1 \cdots q_{(n+3)/2}) dq_1 \cdots dq_{(n+3)/2} \quad (5.23)$$

using the proof of the estimate for G_{κ}' .

The vertex functions are $w_{\kappa}^C(k_i, p_{1,i}, p_{2,i})$ (now without the factor $\tilde{g}$) multiplied by either

$$1, \mu(k_i)^{\delta} (\mu(k_i) + \omega(p_{1,i}) + \omega(p_{2,i}))^{-1} (\delta = 0, 1)$$

or

$$(\omega(p_{1,i}) + \omega(p_{2,i}))(\mu(k_i) + \omega(p_{1,i}) + \omega(p_{2,i}))^{-1},$$

so the estimates

$$\begin{aligned} \mu(k_i)^{-1/2} &= \mu(l_i - p_{1,i} - p_{2,i})^{-1/2} \leq \text{const } \mu(p_{1,i} + p_{2,i})^{-1/2} \mu(l_i)^{1/2}, \\ (\mu(k_i) + \omega(p_{1,i}) + \omega(p_{2,i}))^{-1} &\leq \text{const } \mu(p_{1,i} + p_{2,i}) + \omega(p_{1,i}) + \omega(p_{2,i})^{-1} \mu(l_i), \\ \mu(k_i)(\mu(k_i) + \omega(p_{1,i}) + \omega(p_{2,i}))^{-1} &\leq \text{const}, \\ |\chi_{\kappa}(k_i, p_{1,i}, p_{2,i})| &\leq \text{const}, \end{aligned}$$

and similarly for the derivatives with respect to l_i show that there exists $K' > 0$ with

$$\begin{aligned} |D^{\alpha(n+1)} G_{\kappa}'(l_1 \cdots l_{n+1})| &\leq \prod_{i=1}^{n+1} \mu(l_i)^{3/2} (K')^{n+1} \\ (\alpha(n+1) &= (\alpha_1 \cdots \alpha_{n+1}), \alpha_i \text{ bounded}) \end{aligned} \quad (5.24)$$

uniformly in κ, n and all vacuum diagrams G of order $n+1$.

Setting

$$G_{\kappa}''(l_1 \cdots l_n) = G_{\kappa}'\left(l_1 \cdots l_n, -\sum_{i=1}^n l_i\right) \quad (5.25)$$

we have in particular $G_{\kappa}'' \in O_{\mathcal{M}}(\mathbb{R}^n)$. More precisely $D^{\alpha(n)} \prod_{i=1}^n \mu(l_i)^{-6} G_{\kappa}'' \in L^1(\mathbb{R}^n)$ for all $\alpha(n)$ such that for some $K'' > 0$,

$$\left\| D^{\alpha(n)} \prod_{i=1}^n \mu(l_i)^{-6} G_{\kappa}'' \right\|_1 \leq (K'')^{n+1}, \quad (5.26)$$

$(\alpha(n) = (\alpha_1 \cdots \alpha_n), \alpha_i \text{ bounded})$ uniformly in κ, n and all vacuum diagrams G of order $(n+1)$. Let $G_\kappa(x_1 \cdots x_n)$ be the Fourier transform of $\prod_{i=1}^n \mu(l_i)^{-6} G''_\kappa$. Then the combination of (5.12), (5.23) and (5.26) gives

$$G = \int \left\{ \prod_{i=1}^n \left(m_b^2 - \frac{\partial^2}{\partial x_i^2} \right)^3 g(x_i) \right\} \\ \times g(x_{n+1}) G_\kappa(x_1 - x_{n+1}, \cdots, x_n - x_{n+1}) dx_1 \cdots dx_{n+1}.$$

Using the assumption $g \in \mathfrak{G}(\alpha)$, this gives the estimate

$$\begin{aligned} |G| &\leq \int g(x) dx ((m_b + 1)^3 \alpha)^n \int |G_\kappa(x_1 \cdots x_n)| dx_1 \cdots dx_n \\ &\leq m(g)((m_b + 1)^3 \alpha)^n \left(\int \frac{1}{1+x^2} dx \right)^n \cdot \sup_{(x_1 \cdots x_n) \in \mathbb{R}^n} \left| G_\kappa(x_1 \cdots x_n) \prod_{i=1}^n (1+x_i^2) \right| \\ &\leq m(g)((m_b + 1)^3 \alpha)^n \left(\int \frac{1}{1+x^2} dx \right)^n \\ &\quad \times \int \left| \prod_{i=1}^n \left(1 - \frac{\partial^2}{\partial l_i^2} \right) \mu(l_i)^{-6} G''_\kappa(l_1 \cdots l_n) \right| dl_1 \cdots dl_n \\ &\leq m(g)(K''')^{n+1} \end{aligned} \quad (5.27)$$

for some $K''' = K'''(\alpha) > 0$ uniformly in κ, n and all vacuum diagrams G of order $n+1$, using the estimate (5.26).

Combining this estimate with the estimate on the number of vacuum diagrams contributing to the sum (5.8a), we obtain

$$\begin{aligned} &|\langle \exp A(g, \kappa) H(g, \kappa) \exp -A(g, \kappa) \rangle_0| \\ &\leq m(g) \sum_{\substack{n=2 \\ n \text{ even}}}^{\infty} \frac{(K''')^{n+1} 8(n+1)(500)^n (n+1)!!}{n!} \\ &\leq dm(g) \end{aligned} \quad (5.28)$$

for some $d = d(\alpha)$ uniformly in κ for all $g \in \mathfrak{G}(\alpha)$.

By the above remarks, this proves the upper bound (1.5). Due to property (5.8b) the same estimate holds if we add the boson self-interaction term $:P(\varphi)(g):$ to the Hamiltonian.

6. THE PHYSICAL REPRESENTATION

Corollary 1.2 permits us to derive the same results for the Yukawa theory as were obtained by J. Glimm and A. Jaffe for the $(\varphi^4)_2$ theory [8].

Since essentially no new techniques are required, we just give the results. Let $B \subset \mathbb{R}$ be any bounded open set and $\mathfrak{A}(B)$ the corresponding local (von Neumann) field algebra [10]. In a canonical way

$$\mathfrak{A}(B) \cong \mathfrak{A}_b(B) \times \mathfrak{A}_f(B), \quad (6.1)$$

where $\mathfrak{A}_b(B)$ [resp. $\mathfrak{A}_f(B)$] is the (von Neumann) field algebra of the region B generated by the boson (resp. fermion) field operators alone. Since $\mathfrak{A}_b(B)$ is a factor of type III and $\mathfrak{A}_f(B)$ is a factor [1, 2], by (6.1) $\mathfrak{A}(B)$ is also a factor of type III. This follows from a theorem of Sakai [15], by which the tensor product of two factors is a factor of type III, provided at least one of the factors is such. In particular $\mathfrak{A}(B)$ is simple as a C^* -algebra, since a type III factor contains no nontrivial two-sided ideals.

Define $\mathfrak{A}$ to be the inductive limit of all the C^* -algebras $\mathfrak{A}(B)$ (B open, bounded). Then $\mathfrak{A}$ is also simple.

Let $a \rightarrow V(a)$ be the unitary representation of the translation group. Then the finite propagation speed of $H(g)$ and the relation

$$V(a)^{-1} H(g) V(a) = H(g_a) \quad (g_a(x) = g(x - a)) \quad (6.2)$$

easily give the following

PROPOSITION 6.1. *There exists a unique homomorphism $(a, \tau) \rightarrow \sigma_{a, \tau}$ of the space-time translation group into the group of automorphisms of $\mathfrak{A}$, which is described on $\mathfrak{A}(B)$ by*

$$\sigma_{a, \tau}(A) = V(a)^{-1} e^{iH(\theta)\tau} A e^{-iH(\theta)\tau} V(a), \quad A \in \mathfrak{A}(B) \quad (6.3)$$

if $g(x) = 1$ on $B_{|a|+|\tau|}$.

Here by definition $B_{|a|+|\tau|}$ is the set of all points in $\mathbb{R}$ having distance less than $|a| + |\tau|$ from B .

Choose $\rho \in C_0^\infty(\mathbb{R})$ such that $\rho \geq 0$ and

$$\int \rho(x) dx = 1 = \frac{1}{n} \int \rho\left(\frac{x}{n}\right) dx. \quad (6.4)$$

Let g_n again be defined by (1.14) and let Φ_n be a ground-state vector of $H(g_n)$. Define the following states ω_n on the C^* -algebra $\mathfrak{A}$:

$$\omega_n(A) = \frac{1}{n} \int (\Phi_n, \sigma_x(A) \Phi_n) \rho\left(\frac{x}{n}\right) dx, \quad A \in \mathfrak{A}. \quad (6.5)$$

Since the states of a C^* -algebra form a w^* -compact set by Tychonov's theorem, there is a w^* -convergent subnet of states ω_{n_j} , i.e., for each $A \in \mathfrak{A}$

$$\omega_{n_j}(A) \rightarrow \omega(A)$$

for some state $\omega \in \mathfrak{A}^*$. By the Gelfand–Naimark–Segal construction this state ω defines the scalar product of a new Hilbert space $\mathfrak{H}_{\text{ren}}$, a vector Ω and a unitary representation of the space–time translation group, leaving Ω invariant. We have the following chain of theorems as in [8] (see Theorems 2.1, 2.2 and 2.3 there).

THEOREM 6.2. *Let ω be a w^* -limit point of the sequence $\omega_n \in \mathfrak{A}^*$. Then there exists a subsequence converging w^* to ω . There exists a separable Hilbert space $\mathfrak{H}_{\text{ren}}$, a a^* -isomorphism π of $\mathfrak{A}$, a continuous unitary representation U_{ren} of the space–time translation group and a vector $\Omega \in \mathfrak{H}_{\text{ren}}$, cyclic for $\pi(\mathfrak{A})$, such that*

$$\omega(A) = (\Omega, \pi(A) \Omega) = (\Omega A_{\text{ren}} \Omega), \quad (6.6)$$

$$U_{\text{ren}}(a, \tau)^{-1} \pi(A) U_{\text{ren}}(a, \tau) = \pi(\sigma_{a, \tau} A), \quad (6.7)$$

$$U_{\text{ren}}(a, \tau) \Omega = \Omega \text{ for all } (a, \tau) \in \mathbb{R}^2. \quad (6.8)$$

Then infinitesimal generator H of the time translation is nonnegative.

Thus with the exception of Lorentz covariance the C^* -algebra $\pi(\mathfrak{A})$ satisfies all Haag–Kastler axioms of quantum field theory [12].

THEOREM 6.3. *Let ω be a w^* -limit point of the sequence $\{\omega_n\}$ and let π be the corresponding representation of $\mathfrak{A}$. Then for every bounded open set B in $\mathbb{R}$, there exists a unitary operator $U_B: \mathfrak{H}_{\text{ren}} \rightarrow \mathfrak{H}$, such that*

$$\pi(A) = U_B A U_B^* \quad A \in \mathfrak{A}(B), \quad (6.9)$$

i.e., ω and the corresponding representation π are locally Fock.

The proof of this theorem rests on Griffin's theorem by which isomorphisms between type III factors are unitarily implementable.

THEOREM 6.4. *For every bounded open set $B \subset \mathbb{R}$ the sequence*

$$\{\omega_n \restriction \mathfrak{A}(B)\} \in \mathfrak{A}(B)^*$$

lies in a norm-compact subset of $\mathfrak{A}(B)^$. Any limit point is normal.*

Theorems 6.2 and 6.3 follow from Proposition 6.1 and Theorem 6.4 (see [8, Chapter 2]).

To prove Theorem 6.4, Corollary 1.2 is crucial. As in [8, Chapter 3], Estimate (1.10) gives for all $\zeta \geq 0$ in $C_0^\infty(\mathbb{R})$

$$\omega_n(N_{\tau,\zeta}) \leq M \quad (\tau < \tfrac{1}{2}), \quad (6.10)$$

where $M = M(\zeta)$ is a translation-invariant constant independent of n . Note that $N_{\tau,\zeta} = N_{\tau,\zeta}^0 + N_{\tau,\zeta}^+ + N_{\tau,\zeta}^-$. We then only need to apply Theorem 4.1 in [8]. This theorem was proved in the case of a field algebra constructed from the boson field alone and with an estimate of the form (6.10) with $N_{\tau,\zeta} = N_{\tau,\zeta}^0$. However, it is not difficult to extend the proof of Theorem 4.1 in [8] to the case where $\mathfrak{A}$ also includes the fermion field operators and where (6.10) is valid.

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